Probability Theory (Undergraduate Version)

Yifan Li

August 2026

Contents

Introduction **5**

1 Counting and Basic Probability **6**
1.1 The Basic Principle of Counting 7
1.2 Permutations 7
1.3 Combinations 8
1.4 The Binomial Theorem 8
1.5 Sample Spaces and Events 9
1.6 Axioms of Probability 10
1.7 Basic Properties of Probability 11
1.8 The Inclusion–Exclusion Identity 13

2 Conditional Probability and Independence **14**
2.1 Conditional Probability 15
2.2 The Multiplication Rule 15
2.3 Independence of Events 16
2.4 The Total Probability Formula 17
2.5 Examples of Total Probability 18

3 Random Variables and Probability Distributions **19**
3.1 Discrete Random Variables 20
3.2 Bernoulli Random Variables 20
3.3 Binomial Random Variables 20
3.4 Poisson Distribution as a Binomial Limit 21
3.5 Geometric Random Variables 22
3.6 An Example with Coin Tosses 22
3.7 Expectation of a Discrete Random Variable 22
3.8 Continuous Random Variables 23
3.9 Uniform Distribution 23
3.10 Exponential Random Variables 24
3.11 Gamma Random Variables 25
3.12 Normal Random Variables 25
3.13 Expectation and Variance 26

4 Expectation, Variance, and Distribution Functions **27**
4.1 Properties of Expectation 28
4.2 Variance 28
4.3 Moments of Common Distributions 29
4.4 Cumulative Distribution Functions 32
4.5 Properties of Distribution Functions 32
4.6 Functions of Random Variables 33
4.7 Monotone Transformations 34
4.8 Inverse Transformation 34

5 Joint Distributions and Independence **35**
5.1 Joint Distribution Functions 36
5.2 Jointly Discrete Random Variables 36
5.3 Independence of Random Variables 36

5.4 Jointly Continuous Random Variables . . . 37
5.5 Expectation of Functions of Two Random Variables . . . 38
5.6 An Example of a Joint Density . . . 38
5.7 An Arrival-Time Example . . . 39
5.8 The Bivariate Normal Distribution . . . 39
5.9 Covariance and Correlation . . . 40

6 Sums, Transformations, and Order Statistics . . . 42
6.1 Sums of Independent Random Variables . . . 43
6.2 Sum of Independent Exponential Random Variables . . . 43
6.3 Sum of Independent Normal Random Variables . . . 43
6.4 A Bernoulli–Normal Mixture . . . 44
6.5 Sums of Independent Discrete Random Variables . . . 44
6.6 Transformations of Two Random Variables . . . 45
6.7 The Box–Muller Transformation . . . 46
6.8 A Gamma–Beta Transformation . . . 47
6.9 Partial Sums of Exponential Random Variables . . . 48
6.10 Order Statistics . . . 49
6.11 Marginal Density of an Order Statistic . . . 50

7 Conditional Distributions and Conditional Expectation . . . 52
7.1 Conditional Distribution for Discrete Random Variables . . . 53
7.2 Conditional Distribution of Independent Poisson Variables . . . 53
7.3 Conditional Density for Continuous Random Variables . . . 54
7.4 An Example of a Conditional Density . . . 55
7.5 The Continuous Total Probability Formula . . . 55
7.6 Conditional Expectation . . . 56
7.7 Law of Total Expectation . . . 56
7.8 Conditional Bivariate Normal Distribution . . . 57

8 Generating Functions and Characteristic Functions . . . 58
8.1 Probability Generating Functions . . . 59
8.2 Properties of Probability Generating Functions . . . 59
8.3 Generating Functions of Independent Sums . . . 60
8.4 Moment Generating Functions . . . 61
8.5 Characteristic Functions . . . 62
8.6 Basic Properties of Characteristic Functions . . . 62
8.7 Uniform Continuity of Characteristic Functions . . . 63
8.8 Characteristic Functions of Independent Sums . . . 64
8.9 Examples of Characteristic Functions . . . 64
8.10 Multivariate Normal Distribution . . . 65
8.11 Covariance Matrix and Independence . . . 66
8.12 Sample Mean and Sample Variance . . . 66
8.13 Chi-Squared Distribution . . . 67

9 Modes of Convergence and Laws of Large Numbers . . . 68
9.1 Convergence in Probability and Almost Sure Convergence . . . 69
9.2 The Subsequence Principle . . . 69
9.3 The Borel–Cantelli Lemma . . . 70
9.4 A Monotone Convergence Example . . . 71

9.5 Jensen's Inequality 72
9.6 An Inequality Between Moments 73
9.7 Strong Law of Large Numbers 73
9.8 Fourth-Moment Estimate 74
9.9 Proof of the Strong Law 74
9.10 Dominated Convergence Theorem 75

Acknowledgements **76**

Introduction

Probability theory is the mathematical study of randomness and uncertainty. Its origins are closely connected with games of chance. In the seventeenth century, questions concerning the fair division of stakes in unfinished games led to the famous correspondence between Blaise Pascal and Pierre de Fermat. Their work established some of the earliest systematic methods for calculating probabilities and marked the beginning of classical probability theory.

The subject developed rapidly in the following centuries. Jacob Bernoulli established an early form of the law of large numbers, revealing that stable statistical behavior can emerge from repeated random experiments. Later, Abraham de Moivre, Pierre-Simon Laplace, and Carl Friedrich Gauss developed fundamental ideas concerning binomial probabilities, normal approximation, and continuous distributions. These developments gradually transformed probability from a collection of techniques for games of chance into a general mathematical framework for studying random phenomena.

A decisive step toward the modern theory came in the twentieth century. Probability was placed on a rigorous axiomatic foundation by Andrey Kolmogorov, who formulated probability measures using the language of sets and measure theory. In this framework, an event is a subset of a sample space, a random variable is a function defined on that space, and probability becomes a measure satisfying a small collection of axioms. This viewpoint provides a unified foundation for both discrete and continuous probability.

The central theme of this document is the progression from elementary counting to the mathematical description of random variables. We begin with counting principles, permutations, combinations, sample spaces, events, and the axioms of probability. Conditional probability and independence are then introduced as tools for describing how random events interact.

The discussion subsequently turns to random variables and their distributions. Important discrete and continuous distributions, including the Bernoulli, binomial, Poisson, geometric, uniform, exponential, gamma, and normal distributions, are studied together with their probability mass or density functions. Expectation, variance, cumulative distribution functions, and transformations of random variables provide the basic quantitative tools for analyzing these distributions.

The later chapters develop joint distributions, marginal and conditional distributions, independence, covariance, correlation, and transformations of several random variables. Sums of independent random variables and convolution are considered, followed by conditional expectation and the laws of total probability and total expectation. Probability generating functions, moment generating functions, and characteristic functions provide another perspective on distributions and their sums. Multivariate normal and chi-squared distributions are also introduced.

Finally, the notes connect these distributional ideas with limiting behavior. Convergence in probability and almost sure convergence, the Borel–Cantelli lemma, Jensen's inequality, dominated convergence, and the strong law of large numbers illustrate a fundamental message of probability theory: although individual outcomes are random, large collections of random variables can exhibit remarkably deterministic behavior.

The purpose of these notes is therefore not merely to collect formulas for computing probabilities. The aim is to organize the basic objects, distributions, transformations, and convergence principles into a coherent framework, moving from finite random experiments toward the general probabilistic structures that appear throughout modern mathematics.

Chapter 1

Counting and Basic Probability

1.1 The Basic Principle of Counting

The basic principle of counting is the starting point for computing the number of possible outcomes of an experiment.

Theorem 1.1.1 (Basic Principle of Counting)**.** Suppose there are n experiments. If experiment 1 can result in m_1 possible outcomes, experiment 2 can result in m_2 possible outcomes, and so on, then together there are

$$m_1 m_2 \cdots m_n$$

possible outcomes.

Example 1.1.2. How many functions can be defined from $\{1, 2, \ldots, n\}$ to $\{0, 1\}$?
For each $i \in \{1, 2, \ldots, n\}$, there are two possible choices:

$$f(i) = 0 \qquad \text{or} \qquad f(i) = 1.$$

Therefore, by the basic principle of counting, the total number of functions is

$$2 \times 2 \times \cdots \times 2 = 2^n.$$

1.2 Permutations

A permutation means a different ordered arrangement of objects.

Example 1.2.1. Choose 4 students consecutively without replacement from 80 students. The number of possible ordered outcomes is

$$80 \times 79 \times 78 \times 77.$$

Theorem 1.2.2. Suppose we have n distinct objects. Then there are

$$n! = n(n-1)(n-2)\cdots 2 \cdot 1$$

permutations.

Example 1.2.3. How many different letter arrangements can be formed from the word

BOY?

Since the three letters are distinct, the number of arrangements is

$$3! = 6.$$

If some objects are identical, the preceding formula must be modified.

Theorem 1.2.4. Suppose there are n objects, of which n_1 are alike, n_2 are alike, ..., and n_r are alike, where

$$n_1 + n_2 + \cdots + n_r = n.$$

Then the number of distinct arrangements is

$$\frac{n!}{n_1! n_2! \cdots n_r!}.$$

Example 1.2.5. Consider the letters in

PEPPER.

There are 6 letters in total, with three P's and two E's. Therefore, the number of different arrangements is

$$\frac{6!}{3!2!}.$$

1.3 Combinations

A combination is a selection of objects for which the order of the selected objects does not matter.

Example 1.3.1. Suppose we choose 4 students from 80 students. The number of possible selections is

$$\binom{80}{4} = \frac{80!}{76!4!}.$$

Theorem 1.3.2. If we choose r objects from a total of n distinct objects at a time, then the number of possible combinations is

$$\binom{n}{r} = \frac{n!}{(n-r)!r!}.$$

Example 1.3.3. A student has to sell 3 books from 6 mathematics books, 7 science books, and 4 economics books.
If all three books are to be on the same subject, the number of choices is

$$\binom{6}{3} + \binom{7}{3} + \binom{4}{3}.$$

If the three books are to be on different subjects, the number of choices is

$$\binom{6}{1}\binom{7}{1}\binom{4}{1}.$$

Example 1.3.4. Consider a lattice path from $A = (0, 0)$ to $B = (4, 3)$, where each step is either one unit to the right or one unit upward.
Every path consists of 4 right steps and 3 upward steps. Therefore, there are

$$\binom{7}{3} = \binom{7}{4}$$

possible paths.

1.4 The Binomial Theorem

Theorem 1.4.1 (Binomial Theorem)**.** For any positive integer n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Equivalently,

$$(x+y)^n = \binom{n}{0} y^n + \binom{n}{1} xy^{n-1} + \cdots + \binom{n}{n} x^n.$$

Proof. We prove the result by induction.
For the base case $n = 1$,

$$(x+y)^1 = x + y,$$

which agrees with

$$\sum_{k=0}^{1} \binom{1}{k} x^k y^{1-k}.$$

Assume that

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Then

$$\begin{aligned}(x+y)^{n+1} &= (x+y)(x+y)^n \\ &= (x+y)\sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} \\ &= \sum_{k=0}^{n} \binom{n}{k} x^{k+1} y^{n-k} + \sum_{k=0}^{n} \binom{n}{k} x^k y^{n+1-k}.\end{aligned}$$

Using

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1},$$

we obtain

$$(x+y)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} x^k y^{n+1-k}.$$

Hence the result holds for $n+1$, and therefore for every positive integer n. □

The identity used in the proof can also be verified directly:

$$\begin{aligned}\binom{n}{k} + \binom{n}{k-1} &= \frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!} \\ &= \frac{(n+1)!}{k!(n+1-k)!} \\ &= \binom{n+1}{k}.\end{aligned}$$

1.5 Sample Spaces and Events

Probability is described in terms of three basic objects:

sample space, events, probability measure.

Definition 1.5.1 (Sample Space)**.** The sample space S is the set of all possible outcomes of an experiment.

Example 1.5.2. If we toss a coin once, then

$$S = \{\text{head}, \text{tail}\}.$$

Example 1.5.3. Choose one student uniformly at random from a class of 80 students. If each student is represented by an ordered pair consisting of the student's index and another recorded attribute, the sample space can be written in the corresponding finite form.

Definition 1.5.4 (Event)**.** An event is a subset of the sample space S. Thus, an event E satisfies $E \subseteq S$.

Let E and F be two events.
Their union is

$$E \cup F = \{x : x \in E \text{ or } x \in F\}.$$

Their intersection is

$$E \cap F = \{x : x \in E \text{ and } x \in F\}.$$

The complement of E is

$$E^c = \{x \in S : x \notin E\}.$$

The difference between E and F is

$$E - F = E \cap F^c = \{x : x \in E \text{ and } x \notin F\}.$$

Two events E and F are mutually exclusive if

$$E \cap F = \varnothing.$$

For a sequence of events $E_1, E_2, \ldots$, define

$$\bigcup_{n=1}^{\infty} E_n = \{x : \text{there exists } n \geq 1 \text{ such that } x \in E_n\},$$

and

$$\bigcap_{n=1}^{\infty} E_n = \{x : x \in E_n \text{ for every } n \geq 1\}.$$

Theorem 1.5.5 (De Morgan's Laws)**.** For a sequence of events $E_1, E_2, \ldots$,

$$\left(\bigcup_{n=1}^{\infty} E_n\right)^c = \bigcap_{n=1}^{\infty} E_n^c,$$

and

$$\left(\bigcap_{n=1}^{\infty} E_n\right)^c = \bigcup_{n=1}^{\infty} E_n^c.$$

Proof. We prove the first identity. For any x,

$$\begin{aligned} x \in \left(\bigcup_{n=1}^{\infty} E_n\right)^c &\iff x \notin \bigcup_{n=1}^{\infty} E_n \\ &\iff x \notin E_n \quad \text{for every } n \\ &\iff x \in E_n^c \quad \text{for every } n \\ &\iff x \in \bigcap_{n=1}^{\infty} E_n^c. \end{aligned}$$

The second identity follows similarly. □

1.6 Axioms of Probability

Let S be a sample space.

Definition 1.6.1 (Probability Measure)**.** For each event $E \subseteq S$, the probability $\mathbb{P}(E)$ satisfies the following axioms:

1. $0 \leq \mathbb{P}(E) \leq 1$.
2. $\mathbb{P}(S) = 1$.
3. If $E_1, E_2, \ldots$ are mutually exclusive events, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(E_n).$$

If all outcomes in a finite sample space S are equally likely to occur, then for any event E,

$$\mathbb{P}(E) = \frac{\text{number of outcomes in } E}{\text{number of outcomes in } S}.$$

Example 1.6.2. Suppose there are 365 possible birthdays and birthdays are equally likely. In a class of 80 students, the probability that at least two students have the same birthday is

$$1 - \frac{365 \cdot 364 \cdots (365 - 79)}{365^{80}}.$$

Numerically, this probability is approximately

$$0.999994437.$$

Example 1.6.3 (Poker)**.** A standard deck contains 52 cards. A poker hand consists of 5 cards, so the total number of possible hands is

$$\binom{52}{5}.$$

The probability of a flush is

$$\frac{4\binom{13}{5}}{\binom{52}{5}}.$$

The probability of a straight is

$$\frac{10 \cdot 4^5}{\binom{52}{5}}.$$

The probability of a straight flush is

$$\frac{10 \cdot 4}{\binom{52}{5}}.$$

The probability of a royal flush is

$$\frac{4}{\binom{52}{5}}.$$

1.7 Basic Properties of Probability

The axioms of probability imply several useful properties.

Proposition 1.7.1. For every event E,

$$\mathbb{P}(E^c) = 1 - \mathbb{P}(E).$$

Proof. Since

$$S = E \cup E^c$$

and $E \cap E^c = \varnothing$, countable additivity gives

$$1 = \mathbb{P}(S) = \mathbb{P}(E) + \mathbb{P}(E^c).$$

Therefore,

$$\mathbb{P}(E^c) = 1 - \mathbb{P}(E).$$

□

Proposition 1.7.2. If $E \subseteq F$, then

$$\mathbb{P}(E) \leq \mathbb{P}(F).$$

Proof. Since $E \subseteq F$,

$$F = E \cup (F - E),$$

where the two events on the right-hand side are mutually exclusive. Therefore,

$$\mathbb{P}(F) = \mathbb{P}(E) + \mathbb{P}(F - E) \geq \mathbb{P}(E).$$

□

Proposition 1.7.3. For any two events E and F,

$$\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F).$$

Proof. Write

$$E \cup F = E \cup (F - E).$$

Since the union is disjoint,

$$\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F - E).$$

Also,

$$F = (F - E) \cup (E \cap F),$$

so

$$\mathbb{P}(F) = \mathbb{P}(F - E) + \mathbb{P}(E \cap F).$$

Hence

$$\mathbb{P}(F - E) = \mathbb{P}(F) - \mathbb{P}(E \cap F),$$

which gives

$$\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F).$$

□

1.8 The Inclusion–Exclusion Identity

For two events,

$$\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_2) - \mathbb{P}(E_1 \cap E_2).$$

For three events,

$$\begin{aligned}\mathbb{P}(E_1 \cup E_2 \cup E_3) = \mathbb{P}(E_1) + \mathbb{P}(E_2) + \mathbb{P}(E_3) \\ - \mathbb{P}(E_1 \cap E_2) - \mathbb{P}(E_1 \cap E_3) - \mathbb{P}(E_2 \cap E_3) \\ + \mathbb{P}(E_1 \cap E_2 \cap E_3).\end{aligned}$$

More generally,

$$\begin{aligned}\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \sum_i \mathbb{P}(E_i) - \sum_{i<j} \mathbb{P}(E_i \cap E_j) \\ + \sum_{i<j<k} \mathbb{P}(E_i \cap E_j \cap E_k) - \cdots \\ + (-1)^{n+1} \mathbb{P}\left(\bigcap_{i=1}^{n} E_i\right).\end{aligned}$$

Example 1.8.1. Let $S = \mathbb{Z}_{\geq 0}$, and define

$$\mathbb{P}(\{n\}) = \begin{cases} 1, & n = 0, \\ 0, & n \neq 0. \end{cases}$$

Then this probability measure satisfies the axioms of probability.
In particular, it is possible for every event in a sequence $E_1, E_2, \ldots$ to satisfy $\mathbb{P}(E_n) = 0$, while

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} E_n\right) = 1,$$

depending on the choice of the events. This illustrates why countable unions must be handled through the probability axioms rather than through finite intuition alone.

Chapter 2

Conditional Probability and Independence

2.1 Conditional Probability

Conditional probability describes the probability of an event when some additional information is already known.

Definition 2.1.1 (Conditional Probability)**.** For two events E and F such that $\mathbb{P}(E) > 0$, the conditional probability that F occurs given that E has occurred is defined by

$$\mathbb{P}(F \mid E) = \frac{\mathbb{P}(F \cap E)}{\mathbb{P}(E)}.$$

Example 2.1.2. A class contains 80 students, of whom 10 are female. Choose one student uniformly at random and repeat this experiment twice. Given that at least one of the two selected students is male, find the probability that the other one is female.
Let E denote the event that at least one selected student is male, and let F denote the event that one student is male and the other is female. Then

$$\mathbb{P}(F \mid E) = \frac{\mathbb{P}(F \cap E)}{\mathbb{P}(E)}.$$

Since $F \subseteq E$, we have $F \cap E = F$. Hence

$$\mathbb{P}(F \mid E) = \frac{\frac{70}{80}\frac{10}{80} + \frac{10}{80}\frac{70}{80}}{1 - \frac{10}{80}\frac{10}{80}}.$$

If each outcome in a finite sample space is equally likely, then conditional probability can also be computed by regarding the conditioning event as a reduced sample space.

Example 2.1.3. In poker, two cards are drawn from a standard deck. Suppose the cards are equally distributed among

$$A,\ 2,\ 3,\ \ldots,\ 10,\ J,\ Q,\ K.$$

If a hand has a total of 8 diamonds, what is the probability that the hand contains 3 diamonds?
Let

$$E = \{\text{the hand has a total of 8 diamonds}\},$$

and

$$F = \{\text{the hand contains 3 diamonds}\}.$$

Then

$$\mathbb{P}(F \mid E) = \frac{\mathbb{P}(F \cap E)}{\mathbb{P}(E)}.$$

The counting of admissible hands is performed within the reduced sample space determined by E.

2.2 The Multiplication Rule

The definition of conditional probability can be rearranged into a useful multiplication formula.

Theorem 2.2.1 (Multiplication Rule)**.** For two events E and F,

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F \mid E),$$

provided that $\mathbb{P}(E) > 0$.

More generally, for events $E_1, E_2, \ldots, E_n$,

$$\begin{aligned}\mathbb{P}(E_1 \cap E_2 \cap \cdots \cap E_n) = \mathbb{P}(E_1)\mathbb{P}(E_2 \mid E_1) \\ \times \mathbb{P}(E_3 \mid E_1 \cap E_2) \cdots \\ \times \mathbb{P}(E_n \mid E_1 \cap \cdots \cap E_{n-1}).\end{aligned}$$

Example 2.2.2. An urn contains 8 white balls and 7 black balls. The balls are selected one at a time without replacement.
Find the probability that the first 2 balls are black and the next 2 balls are white.
Let B_i denote the event that the i-th selected ball is black, and W_i the event that the i-th selected ball is white. Then

$$\begin{aligned}&\mathbb{P}(B_1 \cap B_2 \cap W_3 \cap W_4) \\ &\quad = \mathbb{P}(B_1)\mathbb{P}(B_2 \mid B_1)\mathbb{P}(W_3 \mid B_1 \cap B_2) \\ &\qquad \times \mathbb{P}(W_4 \mid B_1 \cap B_2 \cap W_3) \\ &\quad = \frac{7}{15}\frac{6}{14}\frac{8}{13}\frac{7}{12}.\end{aligned}$$

Now find the probability that exactly 2 of the first 4 balls are black. One particular ordering, $B_1W_2B_3W_4$, has probability

$$\frac{7}{15}\frac{8}{14}\frac{6}{13}\frac{7}{12}.$$

There are $\binom{4}{2}$ possible positions for the two black balls. Therefore,

$$\mathbb{P}(\text{exactly two black balls}) = \binom{4}{2}\frac{7 \cdot 8 \cdot 6 \cdot 7}{15 \cdot 14 \cdot 13 \cdot 12}.$$

2.3 Independence of Events

Definition 2.3.1 (Independence)**.** Two events E and F are independent if

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F).$$

Otherwise, they are dependent.

If $\mathbb{P}(E) > 0$, independence is equivalent to

$$\mathbb{P}(F \mid E) = \mathbb{P}(F).$$

Thus, knowing that E has occurred does not change the probability of F.
For a sequence of events $E_1, \ldots, E_n$, independence requires that for every collection of distinct indices $i_1, \ldots, i_k$,

$$\mathbb{P}(E_{i_1} \cap \cdots \cap E_{i_k}) = \prod_{j=1}^{k} \mathbb{P}(E_{i_j}).$$

In particular,

$$\mathbb{P}(E_1 \cap \cdots \cap E_n) = \prod_{i=1}^{n} \mathbb{P}(E_i)$$

when the events are independent.

2.4 The Total Probability Formula

Suppose that

$$S = \bigcup_{i=1}^{n} A_i,$$

where $A_1, \ldots, A_n$ are mutually exclusive.

Theorem 2.4.1 (Total Probability Formula)**.** For any event B,

$$\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i).$$

Proof. Since $A_1, \ldots, A_n$ are mutually exclusive and their union is S,

$$B = B \cap S = \bigcup_{i=1}^{n} (B \cap A_i),$$

and the events $B \cap A_i$ are mutually exclusive. Therefore,

$$\begin{aligned} \mathbb{P}(B) &= \sum_{i=1}^{n} \mathbb{P}(B \cap A_i) \\ &= \sum_{i=1}^{n} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i). \end{aligned}$$

□

The same idea gives a formula for conditional probability.

Proposition 2.4.2. Let $A_1, \ldots, A_n$ be mutually exclusive events such that

$$S = \bigcup_{i=1}^{n} A_i.$$

Then for events B and C with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(C \mid B) = \sum_{i=1}^{n} \mathbb{P}(C \mid B \cap A_i)\mathbb{P}(A_i \mid B).$$

Proof. By the definition of conditional probability,

$$\mathbb{P}(C \mid B) = \frac{\mathbb{P}(C \cap B)}{\mathbb{P}(B)}.$$

Since

$$C \cap B = \bigcup_{i=1}^{n} (C \cap B \cap A_i),$$

we have

$$\begin{aligned} \mathbb{P}(C \mid B) &= \frac{\sum_{i=1}^{n} \mathbb{P}(C \cap B \cap A_i)}{\mathbb{P}(B)} \\ &= \sum_{i=1}^{n} \mathbb{P}(C \mid B \cap A_i) \frac{\mathbb{P}(B \cap A_i)}{\mathbb{P}(B)} \\ &= \sum_{i=1}^{n} \mathbb{P}(C \mid B \cap A_i)\mathbb{P}(A_i \mid B). \end{aligned}$$

□

2.5 Examples of Total Probability

Example 2.5.1. An urn contains 8 white balls and 9 black balls. A fair die is rolled, and the number shown on the die determines the number of balls selected randomly from the urn.
Let

$$A_i = \{\text{the die lands on } i\}, \qquad i = 1, \ldots, 6,$$

and let

$$B = \{\text{all selected balls are white}\}.$$

Then

$$\mathbb{P}(A_i) = \frac{1}{6}.$$

Conditioning on the result of the die,

$$\mathbb{P}(B) = \sum_{i=1}^{6} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i).$$

For a given i,

$$\mathbb{P}(B \mid A_i) = \frac{\binom{8}{i}}{\binom{17}{i}},$$

provided $i \leq 8$. Hence

$$\mathbb{P}(B) = \frac{1}{6} \sum_{i=1}^{6} \frac{\binom{8}{i}}{\binom{17}{i}}.$$

Example 2.5.2. Under the same experiment, suppose that all selected balls are white. Find the conditional probability that the die landed on 3.
Let

$$A = \{\text{the die landed on 3}\},$$

and

$$B = \{\text{all selected balls are white}\}.$$

Then

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

By the multiplication rule,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A).$$

Therefore,

$$\mathbb{P}(A \mid B) = \frac{\frac{1}{6}\frac{\binom{8}{3}}{\binom{17}{3}}}{\frac{1}{6}\sum_{i=1}^{6} \frac{\binom{8}{i}}{\binom{17}{i}}}.$$

Chapter 3

Random Variables and Probability Distributions

3.1 Discrete Random Variables

Definition 3.1.1 (Discrete Random Variable)**.** A random variable $X : S \to \mathbb{R}$ is called a discrete random variable if X can take on at most a countable number of possible values.

Example 3.1.2. Suppose 7 students, of whom 4 are male, are considered. Choose 1 student uniformly at random and repeat this experiment 4 times.
Let X be the number of male students chosen. Then X is a discrete random variable taking values in

$$\{0, 1, 2, 3, 4\}.$$

Moreover, for $k \in \{0, 1, 2, 3, 4\}$,

$$\mathbb{P}(X = k) = \binom{4}{k} \left(\frac{4}{7}\right)^k \left(\frac{3}{7}\right)^{4-k}.$$

Definition 3.1.3 (Probability Mass Function)**.** For a discrete random variable X, the probability mass function, abbreviated as pmf, is defined by

$$p_X(m) = \mathbb{P}(X = m).$$

The pmf completely describes the distribution of a discrete random variable.

3.2 Bernoulli Random Variables

Definition 3.2.1 (Bernoulli Random Variable)**.** A random variable X is said to be a Bernoulli random variable with parameter $p \in [0, 1]$ if

$$\mathbb{P}(X = 0) = 1 - p, \qquad \mathbb{P}(X = 1) = p.$$

We write

$$X \sim \text{Bernoulli}(p).$$

A Bernoulli random variable can be used as an indicator of whether an event occurs:

$$X = \begin{cases} 1, & \text{if the event occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

3.3 Binomial Random Variables

Suppose a coin is tossed independently n times and let X be the number of heads.

Definition 3.3.1 (Binomial Random Variable)**.** A random variable X has a binomial distribution with parameters n and p, denoted by

$$X \sim \text{Bin}(n, p),$$

if

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

We can check that the pmf sums to 1:

$$\begin{aligned} \sum_{k=0}^{n} \mathbb{P}(X = k) &= \sum_{k=0}^{n} \binom{n}{k} p^k (1 - p)^{n-k} \\ &= \Big(p + (1 - p)\Big)^n \\ &= 1. \end{aligned}$$

Example 3.3.2. A student named Alice is selected with probability p in each of 10 independent trials. Let X be the number of times Alice is selected. Then

$$X \sim \text{Bin}(10, p).$$

Equivalently, define

$$X_i = \begin{cases} 1, & \text{if Alice is selected in the } i\text{-th trial,} \\ 0, & \text{otherwise.} \end{cases}$$

Then $X_i \sim \text{Bernoulli}(p)$, and

$$X = \sum_{i=1}^{10} X_i.$$

3.4 Poisson Distribution as a Binomial Limit

Let

$$X \sim \text{Bin}\left(n, \frac{\lambda}{n}\right), \qquad \lambda > 0.$$

Then

$$\begin{aligned} \mathbb{P}(X = k) &= \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{n!}{k!(n-k)!} \frac{\lambda^k}{n^k} \left(1 - \frac{\lambda}{n}\right)^{n-k}. \end{aligned}$$

For any fixed k, as $n \to \infty$,

$$\frac{n!}{(n-k)!n^k} \to 1$$

and

$$\left(1 - \frac{\lambda}{n}\right)^{n-k} \to e^{-\lambda}.$$

Therefore,

$$\mathbb{P}(X = k) \to e^{-\lambda} \frac{\lambda^k}{k!}.$$

Definition 3.4.1 (Poisson Random Variable)**.** A random variable X has a Poisson distribution with parameter $\lambda > 0$ if

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

We write

$$X \sim \text{Poisson}(\lambda).$$

Thus, the Poisson distribution appears naturally as a limiting distribution of binomial random variables when the number of trials is large and the success probability is small.

3.5 Geometric Random Variables

Suppose there is probability p of getting a head. Toss the coin until a head shows up. Let X be the number of tosses needed.

Definition 3.5.1 (Geometric Random Variable)**.** A random variable X is called a geometric random variable with parameter p if

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \qquad k = 1, 2, \ldots.$$

We write

$$X \sim \text{Geometric}(p).$$

Indeed, the event $X = k$ means that the first $k-1$ tosses are tails and the k-th toss is a head. Hence

$$\mathbb{P}(X = k) = (1-p)^{k-1}p.$$

3.6 An Example with Coin Tosses

Example 3.6.1. Suppose a coin is tossed n times. Let X represent the difference between the number of heads and the number of tails. Find the pmf of X.
If U denotes the number of heads and V denotes the number of tails, then

$$U + V = n, \qquad X = U - V.$$

Therefore,

$$U = \frac{n+X}{2}.$$

If n is odd, then the possible values of X are

$$-n, -n+2, \ldots, n-2, n.$$

For such k,

$$\mathbb{P}(X = k) = \binom{n}{(n+k)/2} p^{(n+k)/2}(1-p)^{(n-k)/2}.$$

If n is even, then the possible values are

$$-n, -n+2, \ldots, 0, \ldots, n-2, n,$$

and the same formula applies:

$$\mathbb{P}(X = k) = \binom{n}{(n+k)/2} p^{(n+k)/2}(1-p)^{(n-k)/2}.$$

For all other k,

$$\mathbb{P}(X = k) = 0.$$

3.7 Expectation of a Discrete Random Variable

Definition 3.7.1 (Expectation)**.** For a discrete random variable X, the expectation of X is defined by

$$\mathbb{E}[X] = \sum_m m\mathbb{P}(X = m),$$

whenever the sum is well-defined.

More generally, for a function g,

$$\mathbb{E}[g(X)] = \sum_m g(m)\mathbb{P}(X = m),$$

whenever the sum is well-defined.

3.8 Continuous Random Variables

We now turn from discrete random variables to continuous random variables.

Definition 3.8.1 (Probability Density Function)**.** A nonnegative function

$$f : \mathbb{R} \to [0, \infty)$$

is called a probability density function, abbreviated as pdf, if

$$\int_{-\infty}^{\infty} f(x)\, dx = 1.$$

Definition 3.8.2 (Continuous Random Variable)**.** A random variable X is called a continuous random variable if there exists a pdf f_X such that

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\, dx.$$

In particular, taking $b = a$,

$$\mathbb{P}(X = a) = \int_a^a f_X(x)\, dx = 0.$$

Thus, for a continuous random variable, the probability of taking any single prescribed value is zero.

3.9 Uniform Distribution

Definition 3.9.1 (Uniform Random Variable)**.** A random variable X is uniformly distributed on (a, b), denoted by

$$X \sim \text{Unif}(a, b),$$

if its pdf is

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a < x < b, \\ 0, & \text{otherwise.} \end{cases}$$

In particular,

$$X \sim \text{Unif}(0, 1)$$

has pdf

$$f_X(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Example 3.9.2. Let

$$X \sim \text{Unif}(1, 5).$$

Find $\mathbb{P}(X > 3.5)$.
Since the density is

$$f_X(x) = \frac{1}{4}, \qquad 1 < x < 5,$$

we have

$$\begin{aligned} \mathbb{P}(X > 3.5) &= \int_{3.5}^{5} \frac{1}{4}\, dx \\ &= \frac{3}{8}. \end{aligned}$$

3.10 Exponential Random Variables

Definition 3.10.1 (Exponential Random Variable)**.** We say that X is an exponential random variable with parameter $\lambda > 0$, denoted by

$$X \sim \text{Exp}(\lambda),$$

if its pdf is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

For $t \geq 0$,

$$\begin{aligned} \mathbb{P}(X > t) &= \int_t^\infty \lambda e^{-\lambda x}\, dx \\ &= e^{-\lambda t}. \end{aligned}$$

Definition 3.10.2 (Memoryless Property)**.** A nonnegative random variable X is called memoryless if

$$\mathbb{P}(X > t + s \mid X > t) = \mathbb{P}(X > s), \qquad s, t > 0.$$

The exponential distribution is memoryless. Indeed,

$$\begin{aligned} \mathbb{P}(X > t + s \mid X > t) &= \frac{\mathbb{P}(X > t + s)}{\mathbb{P}(X > t)} \\ &= \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} \\ &= e^{-\lambda s} \\ &= \mathbb{P}(X > s). \end{aligned}$$

The distribution of the remaining waiting time, conditional on having already waited for time t, is therefore the same as the original distribution.

Theorem 3.10.3. If X is a continuous memoryless random variable, then

$$X \sim \text{Exp}(\lambda)$$

for some $\lambda > 0$.

Proof. Let

$$g(t) = \log \mathbb{P}(X > t).$$

By the memoryless property,

$$\mathbb{P}(X > t + s) = \mathbb{P}(X > t)\mathbb{P}(X > s).$$

Therefore,

$$g(t + s) = g(t) + g(s).$$

Since g is continuous, it follows that

$$g(t) = -\lambda t$$

for some $\lambda > 0$. Hence

$$\mathbb{P}(X > t) = e^{-\lambda t},$$

which is the survival function of an exponential random variable. □

Example 3.10.4. Let X be a continuous random variable with pdf

$$f_X(x) = \begin{cases} \dfrac{x}{100} e^{-x^2/200}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

For $0 \leq a < b$,

$$\begin{aligned} \mathbb{P}(a \leq X \leq b) &= \int_a^b \frac{x}{100} e^{-x^2/200}\, dx \\ &= e^{-a^2/200} - e^{-b^2/200}. \end{aligned}$$

In particular,

$$\mathbb{P}(50 \leq X \leq 100) = e^{-50^2/200} - e^{-100^2/200}.$$

Also,

$$\mathbb{P}(X > 100) = e^{-100^2/200}.$$

3.11 Gamma Random Variables

Definition 3.11.1 (Gamma Random Variable)**.** We say that X has a gamma distribution with parameters $\alpha, \lambda > 0$ if its pdf is

$$f_X(x) = \begin{cases} \dfrac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, & x > 0, \\ 0, & x \leq 0, \end{cases}$$

where

$$\Gamma(\alpha) = \int_0^\infty e^{-y} y^{\alpha-1}\, dy.$$

For positive integers n,

$$\Gamma(n) = (n-1)!.$$

Indeed,

$$\Gamma(n) = \int_0^\infty y^{n-1} e^{-y}\, dy.$$

Integration by parts gives

$$\Gamma(n) = (n-1)\Gamma(n-1),$$

and hence

$$\Gamma(n) = (n-1)!.$$

If $X_1, \ldots, X_n$ are independent exponential random variables with parameter λ, then

$$X_1 + \cdots + X_n$$

has a gamma distribution with parameters n and λ.

3.12 Normal Random Variables

Definition 3.12.1 (Normal Random Variable)**.** A random variable X has a normal distribution with mean μ and variance σ^2, denoted by

$$X \sim N(\mu, \sigma^2),$$

if its density is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty.$$

To check that this is a probability density, it is enough to verify the Gaussian integral

$$\int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2\pi}.$$

Let

$$I = \int_{-\infty}^{\infty} e^{-x^2/2}\,dx.$$

Then

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\,dx\,dy.$$

Using polar coordinates,

$$\begin{aligned} I^2 &= \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2} r\,dr\,d\theta \\ &= 2\pi. \end{aligned}$$

Therefore,

$$I = \sqrt{2\pi}.$$

When $\mu = 0$ and $\sigma^2 = 1$, we say that

$$X \sim N(0,1)$$

is a standard normal random variable.

3.13 Expectation and Variance

For a continuous random variable X with pdf f_X, the expectation of X is defined by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx,$$

provided that the integral exists.
More generally, for a function g,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx.$$

For a discrete random variable,

$$\mathbb{E}[X] = \sum_m m\mathbb{P}(X = m),$$

and

$$\mathbb{E}[g(X)] = \sum_m g(m)\mathbb{P}(X = m).$$

Definition 3.13.1 (Variance)**.** The variance of a random variable X is

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right].$$

Equivalently,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Chapter 4

Expectation, Variance, and Distribution Functions

4.1 Properties of Expectation

Recall that for a discrete random variable X,

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x),$$

while for a continuous random variable X with density f_X,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

More generally, if g is a function, then for a discrete random variable,

$$\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x),$$

and for a continuous random variable,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx.$$

Proposition 4.1.1 (Linearity of Expectation)**.** For constants $a, b \in \mathbb{R}$,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

Proof. For a discrete random variable,

$$\begin{aligned}\mathbb{E}[aX + b] &= \sum_x (ax + b)\mathbb{P}(X = x) \\ &= a\sum_x x\mathbb{P}(X = x) + b\sum_x \mathbb{P}(X = x) \\ &= a\mathbb{E}[X] + b.\end{aligned}$$

The continuous case follows similarly:

$$\begin{aligned}\mathbb{E}[aX + b] &= \int_{-\infty}^{\infty} (ax + b) f_X(x)\, dx \\ &= a\mathbb{E}[X] + b.\end{aligned}$$

□

4.2 Variance

Recall that

$$\operatorname{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right].$$

Expanding the square gives

$$\begin{aligned}\operatorname{Var}(X) &= \mathbb{E}\left[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2\right] \\ &= \mathbb{E}[X^2] - 2(\mathbb{E}[X])^2 + (\mathbb{E}[X])^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2.\end{aligned}$$

Proposition 4.2.1. For constants $a, b \in \mathbb{R}$,

$$\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

Proof. Since $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$,

$$\begin{aligned}\operatorname{Var}(aX + b) &= \mathbb{E}\left[\Big(aX + b - a\mathbb{E}[X] - b\Big)^2\right] \\ &= \mathbb{E}\left[a^2(X - \mathbb{E}[X])^2\right] \\ &= a^2 \operatorname{Var}(X).\end{aligned}$$

□

4.3 Moments of Common Distributions

We now compute the expectation and variance of several common distributions.

4.3.1 Bernoulli Distribution

Let $X \sim \text{Bernoulli}(p)$. Then

$$\begin{aligned}\mathbb{E}[X] &= 0 \cdot (1-p) + 1 \cdot p \\ &= p.\end{aligned}$$

Since $X^2 = X$,

$$\mathbb{E}[X^2] = p.$$

Therefore,

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= p - p^2 \\ &= p(1-p).\end{aligned}$$

4.3.2 Binomial Distribution

Let $X \sim \text{Bin}(n, p)$. Then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Thus,

$$\mathbb{E}[X] = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}.$$

Using

$$k\binom{n}{k} = n\binom{n-1}{k-1},$$

we obtain

$$\begin{aligned}\mathbb{E}[X] &= np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \\ &= np.\end{aligned}$$

To compute the second moment, first consider

$$\mathbb{E}[X(X-1)].$$

We have

$$\mathbb{E}[X(X-1)] = \sum_{k=0}^{n} k(k-1) \binom{n}{k} p^k (1-p)^{n-k}.$$

Since

$$k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2},$$

it follows that

$$\mathbb{E}[X(X-1)] = n(n-1)p^2.$$

Since $X^2 = X(X-1) + X$,

$$\mathbb{E}[X^2] = n(n-1)p^2 + np.$$

Therefore,

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= n(n-1)p^2 + np - n^2p^2 \\ &= np(1-p).\end{aligned}$$

Hence, for $X \sim \mathrm{Bin}(n,p)$,

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

4.3.3 Geometric Distribution

Let $X \sim \mathrm{Geometric}(p)$, so

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \qquad k = 1, 2, \ldots.$$

Then

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k(1-p)^{k-1}p.$$

Recall that for $|x| < 1$,

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}.$$

Differentiating gives

$$\sum_{k=1}^{\infty} kx^{k-1} = \frac{1}{(1-x)^2}.$$

Taking $x = 1 - p$,

$$\begin{aligned}\mathbb{E}[X] &= p\frac{1}{p^2} \\ &= \frac{1}{p}.\end{aligned}$$

Similarly,

$$\mathrm{Var}(X) = \frac{1-p}{p^2}.$$

4.3.4 Poisson Distribution

Let $X \sim \mathrm{Poisson}(\lambda)$. Then

$$\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

Therefore,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{k=0}^{\infty} ke^{-\lambda}\frac{\lambda^k}{k!} \\ &= \lambda e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} \\ &= \lambda.\end{aligned}$$

Also,

$$\begin{aligned}\mathbb{E}[X(X-1)] &= \sum_{k=0}^{\infty} k(k-1)e^{-\lambda}\frac{\lambda^k}{k!} \\ &= \lambda^2.\end{aligned}$$

Hence

$$\mathbb{E}[X^2] = \lambda^2 + \lambda,$$

and therefore

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \lambda.\end{aligned}$$

Thus, for a Poisson random variable,

$$\mathbb{E}[X] = \mathrm{Var}(X) = \lambda.$$

4.3.5 Uniform Distribution

Let $X \sim \mathrm{Unif}(a, b)$. Then

$$\begin{aligned}\mathbb{E}[X] &= \int_a^b \frac{x}{b-a}\,dx \\ &= \frac{a+b}{2}.\end{aligned}$$

Also,

$$\begin{aligned}\mathbb{E}[X^2] &= \int_a^b \frac{x^2}{b-a}\,dx \\ &= \frac{a^2+ab+b^2}{3}.\end{aligned}$$

Therefore,

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \frac{(b-a)^2}{12}.\end{aligned}$$

4.3.6 Exponential Distribution

Let $X \sim \mathrm{Exp}(\lambda)$. Then

$$\begin{aligned}\mathbb{E}[X] &= \int_0^\infty x\lambda e^{-\lambda x}\,dx \\ &= \frac{1}{\lambda}.\end{aligned}$$

Similarly,

$$\mathbb{E}[X^2] = \frac{2}{\lambda^2}.$$

Therefore,

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \frac{1}{\lambda^2}.\end{aligned}$$

4.3.7 Normal Distribution

Let $X \sim N(\mu, \sigma^2)$. Then

$$\mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2.$$

In particular, if $Z \sim N(0, 1)$, then

$$\mathbb{E}[Z] = 0, \qquad \mathrm{Var}(Z) = 1.$$

Moreover, if $Z \sim N(0, 1)$, then

$$X = \mu + \sigma Z$$

has distribution $N(\mu, \sigma^2)$.

4.4 Cumulative Distribution Functions

Definition 4.4.1 (Cumulative Distribution Function)**.** The cumulative distribution function, abbreviated as cdf, of a random variable X is defined by

$$F_X(x) = \mathbb{P}(X \leq x), \qquad x \in \mathbb{R}.$$

The cdf can be defined for both discrete and continuous random variables.

Example 4.4.2. Let $X \sim \text{Bernoulli}(p)$. Then

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1-p, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

Example 4.4.3. Let $X \sim \text{Unif}(0,1)$. Then

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ x, & 0 < x < 1, \\ 1, & x \geq 1. \end{cases}$$

Example 4.4.4. Let $X \sim \text{Exp}(\lambda)$. For $x > 0$,

$$\begin{aligned} F_X(x) &= \mathbb{P}(X \leq x) \\ &= \int_0^x \lambda e^{-\lambda t}\, dt \\ &= 1 - e^{-\lambda x}. \end{aligned}$$

Hence

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ 1 - e^{-\lambda x}, & x > 0. \end{cases}$$

4.5 Properties of Distribution Functions

Proposition 4.5.1. For every random variable X, its cdf F_X satisfies:

1. F_X is nondecreasing.
2. $$\lim_{x \to -\infty} F_X(x) = 0.$$
3. $$\lim_{x \to \infty} F_X(x) = 1.$$
4. F_X is right-continuous.

If $a < b$, then

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

For a discrete random variable,

$$\mathbb{P}(X = x) = F_X(x) - F_X(x^-),$$

where

$$F_X(x^-) = \lim_{y \uparrow x} F_X(y).$$

Thus, the jump of the cdf at x is exactly the probability mass at x.
For a continuous random variable with density f_X,

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt.$$

At every point where F_X is differentiable,

$$f_X(x) = F_X'(x).$$

Therefore, the cdf provides a unified description of both discrete and continuous random variables.

4.6 Functions of Random Variables

Suppose X is a random variable and

$$Y = g(X).$$

The distribution of Y can often be found by first computing its cdf:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y).$$

Example 4.6.1. Let $X \sim \text{Unif}(0, 1)$ and define $Y = X^2$.
For $0 < y < 1$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) \\ &= \mathbb{P}(X^2 \leq y) \\ &= \mathbb{P}(X \leq \sqrt{y}) \\ &= \sqrt{y}. \end{aligned}$$

Hence

$$F_Y(y) = \begin{cases} 0, & y \leq 0, \\ \sqrt{y}, & 0 < y < 1, \\ 1, & y \geq 1. \end{cases}$$

Differentiating gives

$$f_Y(y) = \frac{1}{2\sqrt{y}}, \qquad 0 < y < 1.$$

Example 4.6.2. Let $X \sim \text{Unif}(-1, 1)$ and define $Y = X^2$. For $0 < y < 1$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(X^2 \leq y) \\ &= \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= \sqrt{y}. \end{aligned}$$

Therefore,

$$f_Y(y) = \frac{1}{2\sqrt{y}}, \qquad 0 < y < 1.$$

4.7 Monotone Transformations

Suppose X is a continuous random variable with cdf F_X, and let $Y = g(X)$, where g is strictly increasing.
Then

$$\begin{aligned} F_Y(y) &= \mathbb{P}(g(X) \leq y) \\ &= \mathbb{P}(X \leq g^{-1}(y)) \\ &= F_X(g^{-1}(y)). \end{aligned}$$

Differentiating gives

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

If g is strictly decreasing, then

$$\begin{aligned} F_Y(y) &= \mathbb{P}(g(X) \leq y) \\ &= \mathbb{P}(X \geq g^{-1}(y)). \end{aligned}$$

For a continuous random variable,

$$F_Y(y) = 1 - F_X(g^{-1}(y)),$$

and again

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

4.8 Inverse Transformation

The cdf can also be used to construct a random variable with a desired distribution.
Suppose $U \sim \text{Unif}(0, 1)$, and let F be a continuous and strictly increasing cdf. Define

$$X = F^{-1}(U).$$

Then

$$\begin{aligned} \mathbb{P}(X \leq x) &= \mathbb{P}(F^{-1}(U) \leq x) \\ &= \mathbb{P}(U \leq F(x)) \\ &= F(x). \end{aligned}$$

Therefore, X has cdf F.

Example 4.8.1. Let $U \sim \text{Unif}(0, 1)$. We want to construct an exponential random variable with parameter λ.
For $X \sim \text{Exp}(\lambda)$,

$$F_X(x) = 1 - e^{-\lambda x}, \qquad x > 0.$$

Set

$$U = 1 - e^{-\lambda X}.$$

Then

$$e^{-\lambda X} = 1 - U,$$

so

$$X = -\frac{1}{\lambda} \log(1 - U).$$

Since $1 - U \sim \text{Unif}(0, 1)$, we may equivalently use

$$X = -\frac{1}{\lambda} \log U.$$

Thus, a uniform random variable can be transformed into an exponential random variable.

Chapter 5

Joint Distributions and Independence

5.1 Joint Distribution Functions

We now consider two random variables X and Y simultaneously.

Definition 5.1.1 (Joint Cumulative Distribution Function)**.** For random variables X and Y, the joint cumulative distribution function is defined by

$$F_{X,Y}(a,b) = \mathbb{P}(X \leq a,\ Y \leq b), \qquad a,b \in \mathbb{R}.$$

The marginal distribution functions can be recovered from the joint cdf. In particular,

$$\mathbb{P}(X \leq a) = \lim_{b\to\infty} F_{X,Y}(a,b),$$

and similarly,

$$\mathbb{P}(Y \leq b) = \lim_{a\to\infty} F_{X,Y}(a,b).$$

5.2 Jointly Discrete Random Variables

Suppose X and Y are both discrete random variables.

Definition 5.2.1 (Joint Probability Mass Function)**.** The joint probability mass function of X and Y is defined by

$$p_{X,Y}(m,n) = \mathbb{P}(X = m,\ Y = n).$$

The joint cdf is then

$$F_{X,Y}(a,b) = \sum_{m\leq a}\sum_{n\leq b} p_{X,Y}(m,n).$$

The marginal pmfs are obtained by summing out the other variable:

$$p_X(m) = \sum_n p_{X,Y}(m,n),$$

and

$$p_Y(n) = \sum_m p_{X,Y}(m,n).$$

5.3 Independence of Random Variables

Definition 5.3.1 (Independent Random Variables)**.** Random variables X and Y are independent if, for every $A, B \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A,\ Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B).$$

For discrete random variables, X and Y are independent if and only if

$$p_{X,Y}(m,n) = p_X(m)p_Y(n)$$

for all m, n.
Indeed, if X and Y are independent, take $A = \{m\}$ and $B = \{n\}$. Then

$$\begin{aligned} p_{X,Y}(m,n) &= \mathbb{P}(X = m,\ Y = n) \\ &= \mathbb{P}(X = m)\mathbb{P}(Y = n) \\ &= p_X(m)p_Y(n). \end{aligned}$$

Conversely, suppose

$$p_{X,Y}(m,n) = p_X(m)p_Y(n)$$

for all m, n. Then, for any $A, B \subseteq \mathbb{R}$,

$$\begin{aligned}
\mathbb{P}(X \in A,\ Y \in B) &= \sum_{m\in A}\sum_{n\in B} p_{X,Y}(m,n) \\
&= \sum_{m\in A}\sum_{n\in B} p_X(m)p_Y(n) \\
&= \left(\sum_{m\in A} p_X(m)\right)\left(\sum_{n\in B} p_Y(n)\right) \\
&= \mathbb{P}(X \in A)\mathbb{P}(Y \in B).
\end{aligned}$$

Hence X and Y are independent.

5.4 Jointly Continuous Random Variables

Definition 5.4.1 (Joint Probability Density Function)**.** Random variables X and Y are jointly continuous if there exists a nonnegative function $f_{X,Y}(x,y)$ such that, for appropriate sets $A, B \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A,\ Y \in B) = \int_A \int_B f_{X,Y}(x,y)\,dy\,dx.$$

The function $f_{X,Y}$ is called the joint probability density function of X and Y.

The joint cdf is given by

$$F_{X,Y}(a,b) = \int_{-\infty}^{a}\int_{-\infty}^{b} f_{X,Y}(x,y)\,dy\,dx.$$

If the derivatives exist, then

$$f_{X,Y}(a,b) = \frac{\partial^2}{\partial a\,\partial b} F_{X,Y}(a,b).$$

The marginal density functions of X and Y are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy,$$

and

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx.$$

For jointly continuous random variables, X and Y are independent if and only if

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

almost everywhere.

5.5 Expectation of Functions of Two Random Variables

Let $g : \mathbb{R}^2 \to \mathbb{R}$.
If X and Y are discrete, then

$$\mathbb{E}[g(X,Y)] = \sum_m \sum_n g(m,n) p_{X,Y}(m,n).$$

If X and Y are jointly continuous, then

$$\mathbb{E}[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{X,Y}(x,y)\, dy\, dx.$$

For example, if

$$g(X,Y) = \mathbf{1}_{\{X \in A\}} \mathbf{1}_{\{Y \in B\}},$$

then

$$\begin{aligned} \mathbb{E}[g(X,Y)] &= \mathbb{E}\left[\mathbf{1}_{\{X \in A\}} \mathbf{1}_{\{Y \in B\}}\right] \\ &= \mathbb{P}(X \in A,\ Y \in B). \end{aligned}$$

For discrete random variables, this can also be written as

$$\begin{aligned} \mathbb{E}[g(X,Y)] &= \sum_m \sum_n p_{X,Y}(m,n) \mathbf{1}_{\{m \in A\}} \mathbf{1}_{\{n \in B\}} \\ &= \sum_{m \in A} \sum_{n \in B} p_{X,Y}(m,n). \end{aligned}$$

5.6 An Example of a Joint Density

Example 5.6.1. Suppose the joint pdf of X and Y is

$$f_{X,Y}(x,y) = \begin{cases} 2e^{-x-2y}, & x \geq 0,\ y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

First, we check that this is a valid density:

$$\begin{aligned} \int_0^{\infty} \int_0^{\infty} 2e^{-x-2y}\, dy\, dx &= \int_0^{\infty} e^{-x}\, dx \\ &= 1. \end{aligned}$$

For $s > 0$,

$$\begin{aligned} \mathbb{P}(X \leq s) &= \int_0^{s} \int_0^{\infty} 2e^{-x-2y}\, dy\, dx \\ &= \int_0^{s} e^{-x}\, dx \\ &= 1 - e^{-s}. \end{aligned}$$

Similarly,

$$\begin{aligned} \mathbb{P}(Y \leq t) &= \int_0^{t} \int_0^{\infty} 2e^{-x-2y}\, dx\, dy \\ &= 1 - e^{-2t}. \end{aligned}$$

Also,

$$\begin{aligned} \mathbb{P}(X > Y) &= \mathbb{E}[\mathbf{1}_{\{X > Y\}}] \\ &= \int_0^{\infty} \int_y^{\infty} 2e^{-x-2y}\, dx\, dy \\ &= \int_0^{\infty} 2e^{-3y}\, dy \\ &= \frac{2}{3}. \end{aligned}$$

5.7 An Arrival-Time Example

Example 5.7.1. A man and a woman plan to meet at 12:30 P.M. Suppose the man arrives uniformly between 12:15 and 12:45, while the woman arrives uniformly between 12:00 and 1:00. Assume that their arrival times are independent.
After rescaling time so that 12:30 corresponds to 0, let

$$X \sim \text{Unif}\left(-\frac{1}{2}, \frac{1}{2}\right), \qquad Y \sim \text{Unif}(-1, 1).$$

By independence,

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) = \begin{cases} \frac{1}{2}, & -\frac{1}{2} < x < \frac{1}{2}, \quad -1 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

To find the probability that the man arrives first, we compute

$$\begin{aligned} \mathbb{P}(X < Y) &= \iint \mathbf{1}_{\{x<y\}} f_{X,Y}(x,y)\, dx\, dy \\ &= \frac{1}{2} \int_{-1/2}^{1/2} \int_x^1 dy\, dx \\ &= \frac{1}{2} \int_{-1/2}^{1/2} (1-x)\, dx \\ &= \frac{1}{2}. \end{aligned}$$

Now find the probability that the first person to arrive waits no longer than 5 minutes. Under the present scaling, 5 minutes corresponds to 1/6, so we need

$$\mathbb{P}\left(|X - Y| < \frac{1}{6}\right).$$

Hence

$$\begin{aligned} \mathbb{P}\left(|X - Y| < \frac{1}{6}\right) &= \iint \mathbf{1}_{\{|x-y|<1/6\}} f_{X,Y}(x,y)\, dx\, dy \\ &= \frac{1}{2} \int_{-1/2}^{1/2} \int_{x-1/6}^{x+1/6} dy\, dx \\ &= \frac{1}{2} \int_{-1/2}^{1/2} \frac{1}{3}\, dx \\ &= \frac{1}{6}. \end{aligned}$$

5.8 The Bivariate Normal Distribution

The joint pdf of a bivariate normal distribution is

$$\begin{aligned} f_{X,Y}(x,y) = &\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \\ &\times \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}\right]\right\}. \end{aligned}$$

We write

$$(X,Y) \sim N\left(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho\right).$$

The marginal distributions are

$$X \sim N(\mu_X, \sigma_X^2), \qquad Y \sim N(\mu_Y, \sigma_Y^2).$$

Let

$$\boldsymbol{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad \boldsymbol{\mu} = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix},$$

and define the covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}.$$

Then

$$\Sigma^{-1} = \frac{1}{1-\rho^2} \begin{pmatrix} \dfrac{1}{\sigma_X^2} & -\dfrac{\rho}{\sigma_X\sigma_Y} \\ -\dfrac{\rho}{\sigma_X\sigma_Y} & \dfrac{1}{\sigma_Y^2} \end{pmatrix}.$$

The density can therefore be written as

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{2\pi\sqrt{\det\Sigma}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right\}.$$

Indeed,

$$\begin{aligned}
&(\boldsymbol{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) \\
&\quad = \frac{1}{1-\rho^2}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}\right].
\end{aligned}$$

5.9 Covariance and Correlation

Definition 5.9.1 (Covariance)**.** The covariance of X and Y is defined by

$$\mathrm{Cov}(X,Y) = \mathbb{E}\left[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])\right].$$

In particular,

$$\mathrm{Cov}(X,X) = \mathbb{E}\left[(X-\mathbb{E}[X])^2\right] = \mathrm{Var}(X).$$

Expanding the definition gives

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Definition 5.9.2 (Correlation)**.** The correlation coefficient between X and Y is

$$\rho = \mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}.$$

For the bivariate normal distribution,

$$\mathrm{Cov}(X,Y) = \rho\sigma_X\sigma_Y.$$

If X and Y are independent, then

$$\begin{aligned}
\mathrm{Cov}(X,Y) &= \mathbb{E}[(X-\mu_X)(Y-\mu_Y)] \\
&= \mathbb{E}[X-\mu_X]\mathbb{E}[Y-\mu_Y] \\
&= 0.
\end{aligned}$$

Therefore,

$$X, Y \text{ independent} \quad \Longrightarrow \quad \mathrm{Cov}(X, Y) = 0.$$

For a bivariate normal random vector, the converse also holds. Indeed, if $\rho = 0$, then its joint density becomes

$$\begin{aligned} f_{X,Y}(x, y) &= \frac{1}{2\pi\sigma_X\sigma_Y} \exp\left\{-\frac{1}{2}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]\right\} \\ &= \frac{1}{\sqrt{2\pi}\sigma_X} \exp\left(-\frac{(x-\mu_X)^2}{2\sigma_X^2}\right) \frac{1}{\sqrt{2\pi}\sigma_Y} \exp\left(-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}\right) \\ &= f_X(x) f_Y(y). \end{aligned}$$

Hence X and Y are independent.
Thus, for a bivariate normal distribution,

$$X \text{ and } Y \text{ are independent} \quad \Longleftrightarrow \quad \rho = 0.$$

Chapter 6

Sums, Transformations, and Order Statistics

6.1 Sums of Independent Random Variables

Let X and Y be independent continuous random variables with densities f_X and f_Y. Define

$$Z = X + Y.$$

Then

$$\begin{aligned}\mathbb{P}(Z \le z) &= \mathbb{P}(X + Y \le z) \\ &= \iint_{\{x+y\le z\}} f_X(x) f_Y(y)\, dx\, dy.\end{aligned}$$

Writing the region as $x \le z - y$,

$$\begin{aligned}F_Z(z) &= \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_X(x) f_Y(y)\, dx\, dy \\ &= \int_{-\infty}^{\infty} F_X(z-y) f_Y(y)\, dy.\end{aligned}$$

Differentiating with respect to z,

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y)\, dy.$$

Definition 6.1.1 (Convolution)**.** For two functions f and g, their convolution is defined by

$$(f * g)(z) = \int_{-\infty}^{\infty} f(z-y) g(y)\, dy.$$

Thus, if X and Y are independent continuous random variables, then

$$f_{X+Y} = f_X * f_Y.$$

6.2 Sum of Independent Exponential Random Variables

Example 6.2.1. Let $X, Y \sim \text{Exp}(\lambda)$ be independent. Find the density of $X + Y$.
For $z > 0$,

$$\begin{aligned}f_{X+Y}(z) &= \int_{-\infty}^{\infty} f_X(z-y) f_Y(y)\, dy \\ &= \int_0^z \lambda e^{-\lambda(z-y)} \lambda e^{-\lambda y}\, dy \\ &= \lambda^2 e^{-\lambda z} \int_0^z dy \\ &= \lambda^2 z e^{-\lambda z}.\end{aligned}$$

Hence

$$X + Y \sim \text{Gamma}(2, \lambda).$$

More generally, the sum of n independent $\text{Exp}(\lambda)$ random variables has a $\text{Gamma}(n, \lambda)$ distribution.

6.3 Sum of Independent Normal Random Variables

Example 6.3.1. Suppose

$$X \sim N(\mu_X, \sigma_X^2), \qquad Y \sim N(\mu_Y, \sigma_Y^2),$$

and X and Y are independent. Then

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

To see this directly, write

$$\begin{aligned} f_{X+Y}(z) &= \int_{-\infty}^{\infty} f_X(z-y) f_Y(y)\, dy \\ &= \frac{1}{2\pi\sigma_X\sigma_Y} \int_{-\infty}^{\infty} \exp\left\{ -\frac{(z-y-\mu_X)^2}{2\sigma_X^2} - \frac{(y-\mu_Y)^2}{2\sigma_Y^2} \right\} dy. \end{aligned}$$

Completing the square in y gives

$$f_{X+Y}(z) = \frac{1}{\sqrt{2\pi(\sigma_X^2+\sigma_Y^2)}} \exp\left\{ -\frac{(z-\mu_X-\mu_Y)^2}{2(\sigma_X^2+\sigma_Y^2)} \right\}.$$

Therefore,

$$X+Y \sim N(\mu_X+\mu_Y, \sigma_X^2+\sigma_Y^2).$$

6.4 A Bernoulli–Normal Mixture

Example 6.4.1. Suppose

$$X \sim \text{Bernoulli}\left(\frac{1}{2}\right), \qquad Y \sim N(0,1),$$

and X and Y are independent. Find the density of

$$Z = X+Y.$$

Conditioning on the value of X,

$$\mathbb{P}(Z \le z) = \frac{1}{2}\mathbb{P}(Y \le z) + \frac{1}{2}\mathbb{P}(Y \le z-1).$$

Differentiating,

$$f_Z(z) = \frac{1}{2} f_Y(z) + \frac{1}{2} f_Y(z-1).$$

Hence

$$f_Z(z) = \frac{1}{2\sqrt{2\pi}} e^{-z^2/2} + \frac{1}{2\sqrt{2\pi}} e^{-(z-1)^2/2}.$$

6.5 Sums of Independent Discrete Random Variables

For independent discrete random variables X and Y,

$$\begin{aligned} \mathbb{P}(X+Y=n) &= \sum_k \mathbb{P}(X+Y=n \mid X=k)\mathbb{P}(X=k) \\ &= \sum_k \mathbb{P}(Y=n-k)\mathbb{P}(X=k). \end{aligned}$$

Thus,

$$\mathbb{P}(X+Y=n) = \sum_k p_X(k) p_Y(n-k).$$

This is the discrete analogue of convolution.

6.5.1 Poisson Random Variables

Example 6.5.1. Let

$$X \sim \text{Poisson}(\lambda_1), \qquad Y \sim \text{Poisson}(\lambda_2),$$

and suppose X and Y are independent. Then

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2).$$

For $n = 0, 1, 2, \ldots$,

$$\begin{aligned}
\mathbb{P}(X + Y = n) &= \sum_{k=0}^{n} \mathbb{P}(X = k)\mathbb{P}(Y = n - k) \\
&= \sum_{k=0}^{n} e^{-\lambda_1} \frac{\lambda_1^k}{k!} e^{-\lambda_2} \frac{\lambda_2^{n-k}}{(n-k)!} \\
&= e^{-(\lambda_1+\lambda_2)} \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} \lambda_1^k \lambda_2^{n-k} \\
&= e^{-(\lambda_1+\lambda_2)} \frac{(\lambda_1 + \lambda_2)^n}{n!}.
\end{aligned}$$

6.5.2 Binomial Random Variables

Example 6.5.2. Suppose

$$X \sim \text{Bin}(n, p), \qquad Y \sim \text{Bin}(m, p),$$

and X and Y are independent. Then

$$X + Y \sim \text{Bin}(n + m, p).$$

For $0 \leq N \leq n + m$,

$$\begin{aligned}
\mathbb{P}(X + Y = N) &= \sum_{k} \mathbb{P}(X = k)\mathbb{P}(Y = N - k) \\
&= \sum_{k} \binom{n}{k} \binom{m}{N-k} p^N (1-p)^{n+m-N}.
\end{aligned}$$

Using Vandermonde's identity,

$$\sum_{k} \binom{n}{k} \binom{m}{N-k} = \binom{n+m}{N},$$

we obtain

$$\mathbb{P}(X + Y = N) = \binom{n+m}{N} p^N (1-p)^{n+m-N}.$$

6.6 Transformations of Two Random Variables

We now consider a transformation

$$U = g(X, Y), \qquad V = h(X, Y),$$

where X and Y are jointly continuous with joint density $f_{X,Y}$.

Suppose the transformation

$$(x, y) \mapsto (u, v)$$

is one-to-one on the relevant region. Write its inverse as

$$x = r(u, v), \qquad y = s(u, v).$$

Define the Jacobian

$$J(u, v) = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix}.$$

Theorem 6.6.1 (Change of Variables)**.** The joint density of

$$(U, V) = (g(X, Y), h(X, Y))$$

is

$$f_{U,V}(u, v) = f_{X,Y}\Big(r(u, v), s(u, v)\Big)\,|J(u, v)|$$

on the transformed region.

Proof. Let $w : \mathbb{R}^2 \to \mathbb{R}$ be any function. Then

$$\begin{aligned} \mathbb{E}[w(U, V)] &= \mathbb{E}[w(g(X, Y), h(X, Y))] \\ &= \iint w(g(x, y), h(x, y)) f_{X,Y}(x, y)\, dx\, dy. \end{aligned}$$

Using the change of variables

$$x = r(u, v), \qquad y = s(u, v),$$

we obtain

$$\mathbb{E}[w(U, V)] = \iint w(u, v) f_{X,Y}\Big(r(u, v), s(u, v)\Big)\,|J(u, v)|\; du\, dv.$$

On the other hand,

$$\mathbb{E}[w(U, V)] = \iint w(u, v) f_{U,V}(u, v)\, du\, dv.$$

Since w is arbitrary,

$$f_{U,V}(u, v) = f_{X,Y}\Big(r(u, v), s(u, v)\Big)\,|J(u, v)|\,.$$

□

6.7 The Box–Muller Transformation

Example 6.7.1 (Box–Muller Algorithm)**.** Suppose X and Y are independent standard normal random variables. Define

$$R = \sqrt{X^2 + Y^2}, \qquad \Theta = \arctan\left(\frac{Y}{X}\right).$$

Equivalently,

$$X = R\cos\Theta, \qquad Y = R\sin\Theta.$$

The Jacobian is

$$\left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| = r.$$

Since

$$f_{X,Y}(x,y) = \frac{1}{2\pi}e^{-(x^2+y^2)/2},$$

we obtain

$$f_{R,\Theta}(r,\theta) = \frac{1}{2\pi}e^{-r^2/2}r, \qquad r > 0, \quad 0 < \theta < 2\pi.$$

Thus,

$$f_{R,\Theta}(r,\theta) = \left(re^{-r^2/2}\right)\left(\frac{1}{2\pi}\right),$$

so R and Θ are independent.

Now define

$$U = F_R(R).$$

Since

$$F_R(r) = \int_0^r se^{-s^2/2}\,ds = 1 - e^{-r^2/2},$$

we have

$$U = 1 - e^{-R^2/2}.$$

Hence

$$R = \sqrt{-2\log(1-U)}.$$

Since $1 - U \sim \text{Unif}(0,1)$, we can equivalently write

$$R = \sqrt{-2\log U}.$$

If U, V are independent $\text{Unif}(0,1)$ random variables, then

$$R = \sqrt{-2\log U}, \qquad \Theta = 2\pi V.$$

Therefore,

$$X = \sqrt{-2\log U}\cos(2\pi V),$$

and

$$Y = \sqrt{-2\log U}\sin(2\pi V)$$

are independent standard normal random variables.

6.8 A Gamma–Beta Transformation

Example 6.8.1. Let

$$X \sim \text{Gamma}(\alpha,\lambda), \qquad Y \sim \text{Gamma}(\beta,\lambda),$$

and suppose X and Y are independent.
Define

$$U = \frac{X}{X+Y}, \qquad V = X + Y.$$

Find the joint density of U and V.

From

$$U = \frac{X}{X+Y}, \qquad V = X + Y,$$

we obtain

$$X = UV, \qquad Y = V(1-U).$$

The Jacobian is

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \left\| \begin{matrix} v & u \\ -v & 1-u \end{matrix} \right\| = v.$$

By independence,

$$f_{X,Y}(x,y) = f_X(x) f_Y(y).$$

Hence, for $0 < u < 1$ and $v > 0$,

$$\begin{aligned} f_{U,V}(u,v) &= f_{X,Y}(uv, v(1-u))v \\ &= \frac{\lambda^\alpha (uv)^{\alpha-1} e^{-\lambda uv}}{\Gamma(\alpha)} \frac{\lambda^\beta \Big(v(1-u)\Big)^{\beta-1} e^{-\lambda v(1-u)}}{\Gamma(\beta)} v \\ &= \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha)\Gamma(\beta)} u^{\alpha-1}(1-u)^{\beta-1} v^{\alpha+\beta-1} e^{-\lambda v}. \end{aligned}$$

We can factor this as

$$\begin{aligned} f_{U,V}(u,v) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} u^{\alpha-1}(1-u)^{\beta-1} \\ \times \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha+\beta)} v^{\alpha+\beta-1} e^{-\lambda v}. \end{aligned}$$

Thus U and V are independent,

$$U \sim \text{Beta}(\alpha, \beta),$$

and

$$V \sim \text{Gamma}(\alpha+\beta, \lambda).$$

6.9 Partial Sums of Exponential Random Variables

Example 6.9.1. Let

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda),$$

and define

$$Y_k = \sum_{i=1}^{k} X_i, \qquad 1 \le k \le n.$$

Find the joint density of

$$(Y_1, \ldots, Y_n).$$

The transformation is

$$\begin{aligned} Y_1 &= X_1, \\ Y_2 &= X_1 + X_2, \\ &\vdots \\ Y_n &= X_1 + \cdots + X_n. \end{aligned}$$

The inverse transformation is

$$\begin{aligned} X_1 &= Y_1, \\ X_2 &= Y_2 - Y_1, \\ &\vdots \\ X_n &= Y_n - Y_{n-1}. \end{aligned}$$

The absolute value of the Jacobian is 1. Since the X_i's are independent,

$$f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = \lambda^n e^{-\lambda(x_1+\cdots+x_n)}$$

for $x_1,\ldots,x_n > 0$.
Therefore,

$$f_{Y_1,\ldots,Y_n}(y_1,\ldots,y_n) = \lambda^n e^{-\lambda y_n}$$

on the region

$$0 < y_1 < y_2 < \cdots < y_n.$$

Thus,

$$f_{Y_1,\ldots,Y_n}(y_1,\ldots,y_n) = \lambda^n e^{-\lambda y_n} \mathbf{1}_{\{0<y_1<\cdots<y_n\}}.$$

To find the marginal density of Y_n, integrate out $y_1,\ldots,y_{n-1}$:

$$f_{Y_n}(y_n) = \lambda^n e^{-\lambda y_n} \int_0^{y_n} \int_0^{y_{n-1}} \cdots \int_0^{y_2} dy_1 \cdots dy_{n-1}.$$

The volume of the region

$$0 < y_1 < \cdots < y_{n-1} < y_n$$

is

$$\frac{y_n^{n-1}}{(n-1)!}.$$

Hence

$$f_{Y_n}(y) = \frac{\lambda^n y^{n-1} e^{-\lambda y}}{(n-1)!}, \qquad y > 0.$$

Therefore,

$$Y_n \sim \text{Gamma}(n, \lambda).$$

6.10 Order Statistics

Let

$$X_1,\ldots,X_n$$

be i.i.d. random variables with common pdf f and cdf F.

Definition 6.10.1 (Order Statistics)**.** The ordered sequence of $X_1,\ldots,X_n$ is denoted by

$$X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}.$$

The random variable $X_{(j)}$ is called the j-th order statistic.

Since the random variables are continuous,

$$\mathbb{P}(X_i = X_j) = 0, \qquad i \ne j.$$

Thus ties occur with probability zero.

Theorem 6.10.2. The joint density of

$$X_{(1)}, \ldots, X_{(n)}$$

is

$$f_{X_{(1)},\ldots,X_{(n)}}(x_1, \ldots, x_n) = n! \prod_{i=1}^{n} f(x_i)$$

whenever

$$x_1 < x_2 < \cdots < x_n,$$

and is 0 otherwise.

Proof. Suppose

$$x_1 < x_2 < \cdots < x_n.$$

The unordered sample

$$X_1, \ldots, X_n$$

can produce this ordering in $n!$ different ways.
For one particular ordering,

$$\mathbb{P}(X_1 \in dx_1, \ldots, X_n \in dx_n) \approx \prod_{i=1}^{n} f(x_i)\, dx_i.$$

Since there are $n!$ permutations producing the same ordered sample,

$$f_{X_{(1)},\ldots,X_{(n)}}(x_1, \ldots, x_n) = n! \prod_{i=1}^{n} f(x_i)$$

on the region $x_1 < \cdots < x_n$. □

Example 6.10.3. Suppose

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Unif}(0, 1).$$

Since $f(x) = 1$ on $(0, 1)$, the joint density of the order statistics is

$$f_{X_{(1)},\ldots,X_{(n)}}(x_1, \ldots, x_n) = n!$$

whenever

$$0 < x_1 < x_2 < \cdots < x_n < 1.$$

6.11 Marginal Density of an Order Statistic

For $1 \leq j \leq n$, the marginal density of $X_{(j)}$ is

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!(n-j)!} F(x)^{j-1} \Big(1 - F(x)\Big)^{n-j} f(x).$$

To understand this formula, consider the event that $X_{(j)}$ lies near x. We need

$$j - 1$$

of the observations to lie below x, one observation to lie near x, and

$$n - j$$

observations to lie above x.
There are

$$\frac{n!}{(j-1)!(n-j)!}$$

ways to assign these roles. Hence

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!(n-j)!} F(x)^{j-1} \Big(1 - F(x)\Big)^{n-j} f(x).$$

Chapter 7

Conditional Distributions and Conditional Expectation

7.1 Conditional Distribution for Discrete Random Variables

Let X and Y be discrete random variables.

Definition 7.1.1 (Conditional Probability Mass Function)**.** The conditional probability mass function of X given $Y = y$ is defined by

$$p_{X|Y}(x \mid y) = \mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)},$$

provided that $\mathbb{P}(Y = y) > 0$.

Equivalently,

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Definition 7.1.2 (Conditional Cumulative Distribution Function)**.** The conditional cumulative distribution function of X given $Y = y$ is

$$F_{X|Y}(x \mid y) = \mathbb{P}(X \leq x \mid Y = y).$$

For discrete random variables,

$$F_{X|Y}(x \mid y) = \sum_{m \leq x} p_{X|Y}(m \mid y).$$

If X and Y are independent, then

$$\begin{aligned} p_{X|Y}(x \mid y) &= \frac{p_{X,Y}(x, y)}{p_Y(y)} \\ &= \frac{p_X(x) p_Y(y)}{p_Y(y)} \\ &= p_X(x). \end{aligned}$$

Similarly,

$$F_{X|Y}(x \mid y) = F_X(x).$$

Thus, conditioning on Y does not change the distribution of X when X and Y are independent.

7.2 Conditional Distribution of Independent Poisson Variables

Example 7.2.1. Suppose

$$X \sim \text{Poisson}(\lambda_1), \qquad Y \sim \text{Poisson}(\lambda_2),$$

and X and Y are independent.
Find the conditional distribution of X given

$$X + Y = n.$$

For $k = 0, 1, \ldots, n$,

$$\begin{aligned} &\mathbb{P}(X = k \mid X + Y = n) \\ &\quad = \frac{\mathbb{P}(X = k, Y = n - k)}{\mathbb{P}(X + Y = n)}. \end{aligned}$$

Since X and Y are independent,

$$\mathbb{P}(X = k, Y = n - k) = e^{-\lambda_1} \frac{\lambda_1^k}{k!} e^{-\lambda_2} \frac{\lambda_2^{n-k}}{(n-k)!}.$$

Also,

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2),$$

so

$$\mathbb{P}(X + Y = n) = e^{-(\lambda_1+\lambda_2)} \frac{(\lambda_1 + \lambda_2)^n}{n!}.$$

Therefore,

$$\mathbb{P}(X = k \mid X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \times \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}.$$

Hence

$$X \mid (X + Y = n) \sim \text{Bin}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right).$$

7.3 Conditional Density for Continuous Random Variables

Let X and Y be jointly continuous random variables with joint density $f_{X,Y}$.

Definition 7.3.1 (Conditional Probability Density Function)**.** The conditional density of X given $Y = y$ is defined by

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)},$$

provided that $f_Y(y) > 0$.

Since

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx,$$

we can also write

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{\int_{-\infty}^{\infty} f_{X,Y}(t, y)\, dt}.$$

The corresponding conditional cdf is

$$F_{X|Y}(x \mid y) = \mathbb{P}(X \leq x \mid Y = y)$$

and satisfies

$$F_{X|Y}(x \mid y) = \int_{-\infty}^{x} f_{X|Y}(t \mid y)\, dt.$$

If X and Y are independent, then

$$f_{X,Y}(x, y) = f_X(x) f_Y(y),$$

so

$$\begin{aligned} f_{X|Y}(x \mid y) &= \frac{f_X(x) f_Y(y)}{f_Y(y)} \\ &= f_X(x). \end{aligned}$$

7.4 An Example of a Conditional Density

Example 7.4.1. Suppose (X, Y) has joint density

$$f_{X,Y}(x,y) = \begin{cases} 2, & (x,y) \in D, \\ 0, & (x,y) \notin D, \end{cases}$$

where

$$D = \{(x,y) : 0 < y < x < 1\}.$$

Find the conditional distribution of X given $Y = y$.

For fixed $y \in (0,1)$, the possible values of x satisfy

$$y < x < 1.$$

The marginal density of Y is

$$\begin{aligned} f_Y(y) &= \int_y^1 2\,dx \\ &= 2(1-y), \qquad 0 < y < 1. \end{aligned}$$

Therefore,

$$f_{X|Y}(x \mid y) = \begin{cases} \dfrac{1}{1-y}, & y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Hence

$$X \mid (Y = y) \sim \text{Unif}(y, 1).$$

For $y < x < 1$,

$$\begin{aligned} \mathbb{P}(X \le x \mid Y = y) &= \int_y^x \frac{1}{1-y}\,dt \\ &= \frac{x-y}{1-y}. \end{aligned}$$

Thus,

$$F_{X|Y}(x \mid y) = \begin{cases} 0, & x \le y, \\ \dfrac{x-y}{1-y}, & y < x < 1, \\ 1, & x \ge 1. \end{cases}$$

7.5 The Continuous Total Probability Formula

For discrete events, the total probability formula expresses an unconditional probability by conditioning on a partition. The same idea extends to continuous random variables. Suppose Y is a continuous random variable with density f_Y. Then for an event B,

$$\mathbb{P}(B) = \int_{-\infty}^{\infty} \mathbb{P}(B \mid Y = y) f_Y(y)\,dy.$$

In particular,

$$\mathbb{P}(X \le x) = \int_{-\infty}^{\infty} \mathbb{P}(X \le x \mid Y = y) f_Y(y)\,dy.$$

Therefore,

$$F_X(x) = \int_{-\infty}^{\infty} F_{X|Y}(x \mid y) f_Y(y)\, dy.$$

Differentiating with respect to x,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x \mid y) f_Y(y)\, dy.$$

Since

$$f_{X|Y}(x \mid y) f_Y(y) = f_{X,Y}(x, y),$$

this agrees with the marginalization formula

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy.$$

7.6 Conditional Expectation

Conditional distributions allow us to define conditional expectations.

Definition 7.6.1 (Conditional Expectation)**.** For jointly continuous random variables X and Y, the conditional expectation of X given $Y = y$ is

$$\mathbb{E}[X \mid Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x \mid y)\, dx,$$

provided that the integral exists.

More generally,

$$\mathbb{E}[g(X) \mid Y = y] = \int_{-\infty}^{\infty} g(x) f_{X|Y}(x \mid y)\, dx.$$

For discrete random variables,

$$\mathbb{E}[X \mid Y = y] = \sum_{x} x\, p_{X|Y}(x \mid y).$$

Thus $\mathbb{E}[X \mid Y = y]$ is a function of y. Consequently,

$$\mathbb{E}[X \mid Y]$$

is itself a random variable.

7.7 Law of Total Expectation

Theorem 7.7.1 (Law of Total Expectation)**.** If the expectations exist, then

$$\mathbb{E}[X] = \mathbb{E}\Big[\mathbb{E}[X \mid Y]\Big].$$

For continuous random variables,

$$\begin{aligned}
\mathbb{E}[X] &= \int_{-\infty}^{\infty} x f_X(x)\, dx \\
&= \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f_{X|Y}(x \mid y) f_Y(y)\, dy \right] dx.
\end{aligned}$$

Changing the order of integration,

$$\begin{aligned}
\mathbb{E}[X] &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x f_{X|Y}(x \mid y)\, dx \right] f_Y(y)\, dy \\
&= \int_{-\infty}^{\infty} \mathbb{E}[X \mid Y = y] f_Y(y)\, dy \\
&= \mathbb{E}\Big[\mathbb{E}[X \mid Y]\Big].
\end{aligned}$$

7.8 Conditional Bivariate Normal Distribution

Recall that if X and Y have a bivariate normal distribution, their joint density is

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \times \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right\}.$$

The marginal density of Y is

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma_Y}\exp\left(-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}\right).$$

Therefore,

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

After simplifying the exponent,

$$f_{X|Y}(x \mid y) = \frac{1}{\sqrt{2\pi}\sigma_X\sqrt{1-\rho^2}} \times \exp\left\{-\frac{\left[x-\mu_X-\rho\frac{\sigma_X}{\sigma_Y}(y-\mu_Y)\right]^2}{2\sigma_X^2(1-\rho^2)}\right\}.$$

Hence,

$$X \mid (Y=y) \sim N\left(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y-\mu_Y),\, \sigma_X^2(1-\rho^2)\right).$$

In particular,

$$\mathbb{E}[X \mid Y=y] = \mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y-\mu_Y),$$

and

$$\mathrm{Var}(X \mid Y=y) = \sigma_X^2(1-\rho^2).$$

If $\rho = 0$, then

$$f_{X|Y}(x \mid y) = f_X(x),$$

which is consistent with the fact that jointly normal random variables with zero correlation are independent.

Chapter 8

Generating Functions and Characteristic Functions

8.1 Probability Generating Functions

Let X be a nonnegative discrete random variable.

Definition 8.1.1 (Probability Generating Function)**.** The probability generating function of X is defined by

$$G_X(s) = \mathbb{E}[s^X] = \sum_{k=0}^{\infty} s^k \mathbb{P}(X = k),$$

whenever the series converges.

The series is absolutely convergent at least for $|s| \leq 1$.

Example 8.1.2 (Bernoulli Distribution)**.** If $X \sim \text{Bernoulli}(p)$, then

$$\begin{aligned} G_X(s) &= (1-p)s^0 + ps^1 \\ &= 1 - p + ps. \end{aligned}$$

Example 8.1.3 (Binomial Distribution)**.** If $X \sim \text{Bin}(n, p)$, then

$$\begin{aligned} G_X(s) &= \sum_{k=0}^{n} s^k \binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=0}^{n} \binom{n}{k} (sp)^k (1-p)^{n-k} \\ &= (1 - p + sp)^n. \end{aligned}$$

Example 8.1.4 (Poisson Distribution)**.** If $X \sim \text{Poisson}(\lambda)$, then

$$\begin{aligned} G_X(s) &= \sum_{k=0}^{\infty} s^k e^{-\lambda} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda s)^k}{k!} \\ &= e^{-\lambda} e^{\lambda s} \\ &= e^{\lambda(s-1)}. \end{aligned}$$

8.2 Properties of Probability Generating Functions

The probability generating function contains information about both the distribution and the moments of a nonnegative discrete random variable.

Proposition 8.2.1. For $k = 0, 1, 2, \ldots$,

$$\mathbb{P}(X = k) = \frac{1}{k!} G_X^{(k)}(0).$$

Proof. Starting from

$$G_X(s) = \sum_{m=0}^{\infty} s^m \mathbb{P}(X = m),$$

differentiate k times:

$$G_X^{(k)}(s) = \sum_{m=k}^{\infty} m(m-1)\cdots(m-k+1) s^{m-k} \mathbb{P}(X = m).$$

Setting $s = 0$, only the term $m = k$ remains. Hence

$$G_X^{(k)}(0) = k!\mathbb{P}(X = k).$$

Therefore,

$$\mathbb{P}(X = k) = \frac{1}{k!}G_X^{(k)}(0).$$

□

Also,

$$G_X(1) = 1.$$

Differentiating once,

$$G'_X(s) = \sum_{m=1}^{\infty} ms^{m-1}\mathbb{P}(X = m).$$

Thus,

$$G'_X(1) = \sum_{m=1}^{\infty} m\mathbb{P}(X = m) = \mathbb{E}[X].$$

Differentiating twice,

$$G''_X(1) = \mathbb{E}[X(X - 1)].$$

Since

$$X^2 = X(X - 1) + X,$$

we obtain

$$\mathbb{E}[X^2] = G''_X(1) + G'_X(1).$$

Therefore,

$$\mathrm{Var}(X) = G''_X(1) + G'_X(1) - \Big(G'_X(1)\Big)^2.$$

8.3 Generating Functions of Independent Sums

Theorem 8.3.1. If $X_1, \ldots, X_n$ are independent nonnegative discrete random variables and

$$Y = X_1 + \cdots + X_n,$$

then

$$G_Y(s) = \prod_{i=1}^{n} G_{X_i}(s).$$

Proof. By independence,

$$\begin{aligned} G_Y(s) &= \mathbb{E}[s^{X_1+\cdots+X_n}] \\ &= \mathbb{E}[s^{X_1} \cdots s^{X_n}] \\ &= \prod_{i=1}^{n} \mathbb{E}[s^{X_i}] \\ &= \prod_{i=1}^{n} G_{X_i}(s). \end{aligned}$$

□

Example 8.3.2. Suppose

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Geometric}(p),$$

where

$$\mathbb{P}(X_i = k) = (1-p)^{k-1}p, \qquad k \geq 1.$$

Then

$$\begin{aligned} G_{X_i}(s) &= \sum_{k=1}^{\infty} s^k (1-p)^{k-1} p \\ &= ps \sum_{k=0}^{\infty} \Big((1-p)s\Big)^k \\ &= \frac{ps}{1-(1-p)s}. \end{aligned}$$

If

$$T = X_1 + \cdots + X_n,$$

then

$$G_T(s) = \left(\frac{ps}{1-(1-p)s}\right)^n.$$

8.4 Moment Generating Functions

Definition 8.4.1 (Moment Generating Function)**.** The moment generating function of a random variable X is defined by

$$M_X(t) = \mathbb{E}[e^{tX}],$$

whenever this expectation is finite.

Using the power series

$$e^{tX} = \sum_{n=0}^{\infty} \frac{t^n X^n}{n!},$$

formally,

$$\begin{aligned} M_X(t) &= \mathbb{E}[e^{tX}] \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbb{E}[X^n]. \end{aligned}$$

Hence, whenever differentiation under the expectation is justified,

$$M_X^{(n)}(0) = \mathbb{E}[X^n].$$

In particular,

$$M_X'(0) = \mathbb{E}[X],$$

and

$$M_X''(0) = \mathbb{E}[X^2].$$

Therefore,

$$\text{Var}(X) = M_X''(0) - \Big(M_X'(0)\Big)^2.$$

However, the moment generating function need not exist for every random variable.

Example 8.4.2. Consider a random variable with density

$$f_X(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Then, for $t < 1$,

$$\begin{aligned} M_X(t) &= \int_0^\infty e^{tx} e^{-x}\, dx \\ &= \int_0^\infty e^{-(1-t)x}\, dx \\ &= \frac{1}{1-t}. \end{aligned}$$

For $t \geq 1$, the integral diverges.

8.5 Characteristic Functions

The characteristic function is defined for every random variable, even when the moment generating function does not exist.

Definition 8.5.1 (Characteristic Function)**.** The characteristic function of a random variable X is

$$\phi_X(t) = \mathbb{E}[e^{itX}], \qquad t \in \mathbb{R}.$$

Using Euler's formula,

$$e^{itX} = \cos(tX) + i \sin(tX),$$

so

$$\phi_X(t) = \mathbb{E}[\cos(tX)] + i\mathbb{E}[\sin(tX)].$$

For a continuous random variable,

$$\phi_X(t) = \int_{-\infty}^\infty e^{itx}\, dF_X(x),$$

and, if a density exists,

$$\phi_X(t) = \int_{-\infty}^\infty e^{itx} f_X(x)\, dx.$$

The distribution of X is uniquely determined by its characteristic function.

8.6 Basic Properties of Characteristic Functions

Proposition 8.6.1. For every random variable X,

$$\phi_X(0) = 1.$$

Indeed,

$$\phi_X(0) = \mathbb{E}[e^{i0X}] = 1.$$

Proposition 8.6.2. For every $t \in \mathbb{R}$,

$$|\phi_X(t)| \leq 1.$$

Proof. Since $\left|e^{itX}\right| = 1$,

$$\begin{aligned} |\phi_X(t)| &= \left|\mathbb{E}[e^{itX}]\right| \\ &\leq \mathbb{E}[\left|e^{itX}\right|] \\ &= 1. \end{aligned}$$

□

Also,

$$\phi_X(-t) = \overline{\phi_X(t)}.$$

Indeed,

$$\begin{aligned} \phi_X(-t) &= \mathbb{E}[e^{-itX}] \\ &= \mathbb{E}[\cos(tX) - i\sin(tX)] \\ &= \overline{\mathbb{E}[\cos(tX) + i\sin(tX)]}. \end{aligned}$$

Proposition 8.6.3. If $\mathbb{E}[|X|^k] < \infty$, then

$$\phi_X^{(k)}(0) = i^k \mathbb{E}[X^k].$$

For example,

$$\phi_X'(0) = i\mathbb{E}[X],$$

and

$$\phi_X''(0) = -\mathbb{E}[X^2].$$

8.7 Uniform Continuity of Characteristic Functions

Theorem 8.7.1. Every characteristic function is uniformly continuous on $\mathbb{R}$.

Proof. For $s, t \in \mathbb{R}$,

$$\begin{aligned} |\phi_X(t+s) - \phi_X(t)| &= \left|\mathbb{E}[e^{itX}(e^{isX} - 1)]\right| \\ &\leq \mathbb{E}[\left|e^{isX} - 1\right|]. \end{aligned}$$

As $s \to 0$,

$$e^{isX} \to 1$$

pointwise, while

$$\left|e^{isX} - 1\right| \leq 2.$$

By the dominated convergence theorem,

$$\mathbb{E}[\left|e^{isX} - 1\right|] \to 0.$$

Hence

$$|\phi_X(t+s) - \phi_X(t)| \to 0$$

uniformly in t. □

8.8 Characteristic Functions of Independent Sums

Theorem 8.8.1. If $X_1, \ldots, X_n$ are independent and

$$Y = X_1 + \cdots + X_n,$$

then

$$\phi_Y(t) = \prod_{j=1}^{n} \phi_{X_j}(t).$$

Proof. By independence,

$$\begin{aligned}
\phi_Y(t) &= \mathbb{E}[e^{it(X_1+\cdots+X_n)}] \\
&= \mathbb{E}[e^{itX_1} \cdots e^{itX_n}] \\
&= \prod_{j=1}^{n} \mathbb{E}[e^{itX_j}] \\
&= \prod_{j=1}^{n} \phi_{X_j}(t).
\end{aligned}$$

□

8.9 Examples of Characteristic Functions

Example 8.9.1 (Bernoulli Distribution)**.** Let $X \sim \text{Bernoulli}(p)$. Then

$$\begin{aligned}
\phi_X(t) &= (1-p)e^{it\cdot 0} + pe^{it} \\
&= 1 - p + pe^{it}.
\end{aligned}$$

Example 8.9.2 (Binomial Distribution)**.** Let $X \sim \text{Bin}(n, p)$. Then

$$\begin{aligned}
\phi_X(t) &= \sum_{k=0}^{n} e^{itk} \binom{n}{k} p^k (1-p)^{n-k} \\
&= \left(1 - p + pe^{it}\right)^n.
\end{aligned}$$

Example 8.9.3 (Poisson Distribution)**.** Let $X \sim \text{Poisson}(\lambda)$. Then

$$\begin{aligned}
\phi_X(t) &= \sum_{k=0}^{\infty} e^{itk} e^{-\lambda} \frac{\lambda^k}{k!} \\
&= e^{-\lambda} \exp(\lambda e^{it}) \\
&= \exp\left\{\lambda(e^{it} - 1)\right\}.
\end{aligned}$$

Example 8.9.4 (Normal Distribution)**.** Let $X \sim N(\mu, \sigma^2)$. Then

$$\phi_X(t) = \exp\left(i\mu t - \frac{1}{2}\sigma^2 t^2\right).$$

To see this, write

$$\phi_X(t) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \exp\left\{itx - \frac{(x-\mu)^2}{2\sigma^2}\right\} dx.$$

Completing the square yields

$$\phi_X(t) = e^{i\mu t - \sigma^2 t^2/2}.$$

8.10 Multivariate Normal Distribution

Let

$$Z_1, \ldots, Z_n \overset{\text{i.i.d.}}{\sim} N(0,1).$$

Then

$$\boldsymbol{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}$$

is called a standard n-dimensional normal random vector.

Definition 8.10.1 (Multivariate Normal Random Vector)**.** A random vector

$$\boldsymbol{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_m \end{pmatrix}$$

is called multivariate normal if it can be written as

$$\boldsymbol{X} = A\boldsymbol{Z} + \boldsymbol{\mu},$$

where A is a deterministic matrix, $\boldsymbol{Z}$ is a standard normal random vector, and $\boldsymbol{\mu} \in \mathbb{R}^m$.

Let

$$\boldsymbol{\mu} = (\mu_1, \ldots, \mu_m)^T$$

and

$$\Sigma = AA^T.$$

Then

$$\Sigma = \Big(\mathrm{Cov}(X_i, X_j)\Big)_{i,j=1}^{m}.$$

The characteristic function of $\boldsymbol{X}$ is

$$\phi_{\boldsymbol{X}}(\boldsymbol{t}) = \mathbb{E}[e^{i\boldsymbol{t}^T\boldsymbol{X}}].$$

Since

$$\boldsymbol{X} = A\boldsymbol{Z} + \boldsymbol{\mu},$$

we have

$$\begin{aligned} \phi_{\boldsymbol{X}}(\boldsymbol{t}) &= \mathbb{E}\left[e^{i\boldsymbol{t}^T(A\boldsymbol{Z}+\boldsymbol{\mu})}\right] \\ &= e^{i\boldsymbol{t}^T\boldsymbol{\mu}}\mathbb{E}\left[e^{i(A^T\boldsymbol{t})^T\boldsymbol{Z}}\right]. \end{aligned}$$

Since the components of $\boldsymbol{Z}$ are independent standard normal random variables,

$$\phi_{\boldsymbol{X}}(\boldsymbol{t}) = \exp\left\{i\boldsymbol{t}^T\boldsymbol{\mu} - \frac{1}{2}\boldsymbol{t}^T\Sigma\boldsymbol{t}\right\}.$$

Thus,

$$\boxed{\phi_{\boldsymbol{X}}(\boldsymbol{t}) = \exp\left\{i\boldsymbol{t}^T\boldsymbol{\mu} - \frac{1}{2}\boldsymbol{t}^T\Sigma\boldsymbol{t}\right\}}.$$

If Σ is positive definite, then the joint density is

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{(2\pi)^{m/2}(\det\Sigma)^{1/2}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right\}.$$

8.11 Covariance Matrix and Independence

For a multivariate normal random vector,

$$\Sigma_{ij} = \mathrm{Cov}(X_i, X_j).$$

Suppose

$$\boldsymbol{X} = A\boldsymbol{Z} + \boldsymbol{\mu}.$$

Writing

$$X_i = \sum_k a_{ik} Z_k + \mu_i,$$

and

$$X_j = \sum_k a_{jk} Z_k + \mu_j,$$

we obtain

$$\begin{aligned} \mathrm{Cov}(X_i, X_j) &= \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] \\ &= \sum_{k,\ell} a_{ik} a_{j\ell} \mathbb{E}[Z_k Z_\ell]. \end{aligned}$$

Since

$$\mathbb{E}[Z_k Z_\ell] = \begin{cases} 1, & k = \ell, \\ 0, & k \neq \ell, \end{cases}$$

it follows that

$$\mathrm{Cov}(X_i, X_j) = \sum_k a_{ik} a_{jk}.$$

Therefore,

$$\Sigma = AA^T.$$

For jointly normal random variables, zero covariance implies independence. In particular, if Σ is diagonal, then

$$X_1, \ldots, X_m$$

are independent.

8.12 Sample Mean and Sample Variance

Suppose

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2).$$

Define the sample mean by

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Then

$$\mathbb{E}[\overline{X}] = \mu.$$

Define the sample variance by

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

Using

$$\sum_{i=1}^{n} (X_i - \overline{X})^2 = \sum_{i=1}^{n} X_i^2 - n\overline{X}^2,$$

we obtain

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \overline{X})^2\right] = n(\sigma^2 + \mu^2) - n\mathbb{E}[\overline{X}^2].$$

Since

$$\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n},$$

we have

$$\mathbb{E}[\overline{X}^2] = \frac{\sigma^2}{n} + \mu^2.$$

Therefore,

$$\begin{aligned}\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \overline{X})^2\right] &= n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) \\ &= (n-1)\sigma^2.\end{aligned}$$

Hence

$$\mathbb{E}[S^2] = \sigma^2.$$

8.13 Chi-Squared Distribution

Let

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(0,1).$$

Definition 8.13.1 (Chi-Squared Random Variable)**.** The random variable

$$Y = X_1^2 + \cdots + X_n^2$$

is said to have a chi-squared distribution with n degrees of freedom. We write

$$Y \sim \chi_n^2.$$

For one standard normal random variable X, let

$$Y = X^2.$$

It is enough to determine the distribution of Y.
For $y > 0$,

$$\begin{aligned}F_Y(y) &= \mathbb{P}(X^2 \le y) \\ &= \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) \\ &= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx.\end{aligned}$$

Differentiating with respect to y,

$$f_Y(y) = \frac{1}{\sqrt{2\pi y}} e^{-y/2}, \qquad y > 0.$$

This is a Gamma distribution:

$$X^2 \sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2}\right).$$

More generally,

$$\chi_n^2 \sim \text{Gamma}\left(\frac{n}{2}, \frac{1}{2}\right),$$

with density

$$f_{\chi_n^2}(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}, \qquad x > 0.$$

Chapter 9

Modes of Convergence and Laws of Large Numbers

9.1 Convergence in Probability and Almost Sure Convergence

We first recall two important modes of convergence for random variables.

Definition 9.1.1 (Convergence in Probability)**.** A sequence of random variables X_n converges in probability to X, written

$$X_n \xrightarrow{P} X,$$

if for every $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \to 0$$

as $n \to \infty$.

Definition 9.1.2 (Almost Sure Convergence)**.** A sequence of random variables X_n converges almost surely to X, written

$$X_n \xrightarrow{\text{a.s.}} X,$$

if

$$\mathbb{P}\left(\lim_{n\to\infty} X_n = X\right) = 1.$$

Almost sure convergence implies convergence in probability:

$$X_n \xrightarrow{\text{a.s.}} X \quad \Longrightarrow \quad X_n \xrightarrow{P} X.$$

The converse is not true in general. However, convergence in probability always contains an almost surely convergent subsequence.

9.2 The Subsequence Principle

Theorem 9.2.1 (Subsequence Characterization)**.** The following are equivalent:

$$X_n \xrightarrow{P} X$$

and

$$\text{for every subsequence } X_{n_k}, \text{ there exists a further subsequence } X_{n_{k(m)}}$$

such that

$$X_{n_{k(m)}} \xrightarrow{\text{a.s.}} X.$$

Proof. Suppose first that every subsequence X_{n_k} has a further subsequence $X_{n_{k(m)}}$ such that

$$X_{n_{k(m)}} \xrightarrow{\text{a.s.}} X.$$

Then

$$X_{n_{k(m)}} \xrightarrow{P} X.$$

Fix $\varepsilon > 0$, and define

$$y_n = \mathbb{P}(|X_n - X| > \varepsilon).$$

If X_n does not converge to X in probability, then there exists $\varepsilon > 0$ such that $y_n \not\to 0$. Hence there exists a subsequence y_{n_k} and some constant $c > 0$ such that

$$y_{n_k} > c$$

for all k.

But by assumption, X_{n_k} has a further subsequence $X_{n_{k(m)}}$ satisfying

$$X_{n_{k(m)}} \xrightarrow{\text{a.s.}} X,$$

and therefore

$$X_{n_{k(m)}} \xrightarrow{P} X.$$

Thus,

$$\mathbb{P}(\left|X_{n_{k(m)}} - X\right| > \varepsilon) \to 0,$$

which contradicts $y_{n_k} > c$.
Therefore,

$$X_n \xrightarrow{P} X.$$

Conversely, suppose

$$X_n \xrightarrow{P} X.$$

Then, for every $k \geq 1$,

$$\mathbb{P}\left(|X_n - X| > \frac{1}{k}\right) \to 0.$$

Hence we may choose a subsequence X_{n_k} such that

$$\mathbb{P}\left(|X_{n_k} - X| > \frac{1}{k}\right) \leq \frac{1}{k^2}.$$

Therefore,

$$\sum_{k=1}^{\infty} \mathbb{P}\left(|X_{n_k} - X| > \frac{1}{k}\right) \leq \sum_{k=1}^{\infty} \frac{1}{k^2} < \infty.$$

By the Borel–Cantelli lemma,

$$\mathbb{P}\left(|X_{n_k} - X| > \frac{1}{k} \text{ infinitely often}\right) = 0.$$

Thus, with probability 1, there exists K such that for all $k \geq K$,

$$|X_{n_k} - X| \leq \frac{1}{k}.$$

Hence

$$X_{n_k} \xrightarrow{\text{a.s.}} X.$$

□

9.3 The Borel–Cantelli Lemma

We now state the form of the Borel–Cantelli lemma used above.

Definition 9.3.1 (Infinitely Often)**.** For a sequence of events $A_1, A_2, \ldots$,

$$\{A_n \text{ infinitely often}\} = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n.$$

The event A_n occurs infinitely often if, for every N, there exists some $n \geq N$ such that A_n occurs.
Its complement is

$$\left(\bigcap_{N=1}^{\infty}\bigcup_{n=N}^{\infty} A_n\right)^c = \bigcup_{N=1}^{\infty}\bigcap_{n=N}^{\infty} A_n^c.$$

Thus, the complement means that there exists N such that for all $n \geq N$, A_n does not occur.

Lemma 9.3.2 (Borel–Cantelli)**.** Let $A_1, A_2, \ldots$ be a sequence of events. If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

then

$$\mathbb{P}(A_n \text{ infinitely often}) = 0.$$

Proof. Let $\mathbf{1}_{A_n}$ be the indicator of A_n. Then

$$\begin{aligned}\mathbb{E}\left[\sum_{n=1}^{\infty} \mathbf{1}_{A_n}\right] &= \sum_{n=1}^{\infty} \mathbb{E}[\mathbf{1}_{A_n}] \\ &= \sum_{n=1}^{\infty} \mathbb{P}(A_n) \\ &< \infty.\end{aligned}$$

Therefore,

$$\mathbb{P}\left(\sum_{n=1}^{\infty} \mathbf{1}_{A_n} < \infty\right) = 1.$$

But

$$\sum_{n=1}^{\infty} \mathbf{1}_{A_n} < \infty$$

means that only finitely many of the events A_n occur, whereas

$$\sum_{n=1}^{\infty} \mathbf{1}_{A_n} = \infty$$

means that A_n occurs infinitely often. Hence

$$\mathbb{P}(A_n \text{ infinitely often}) = 0.$$

□

9.4 A Monotone Convergence Example

Example 9.4.1. Suppose

$$X_1 \geq X_2 \geq X_3 \geq \cdots$$

are nonnegative random variables and

$$X_n \xrightarrow{P} 0.$$

Then

$$X_n \xrightarrow{\text{a.s.}} 0.$$

Proof. Since

$$X_n \xrightarrow{P} 0,$$

the subsequence principle gives a subsequence X_{n_k} such that

$$X_{n_k} \xrightarrow{\text{a.s.}} 0.$$

Thus,

$$\mathbb{P}\left(\lim_{k\to\infty} X_{n_k} = 0\right) = 1.$$

Fix an outcome for which $X_{n_k} \to 0$. For every $\varepsilon > 0$, there exists k_0 such that for $k \geq k_0$,

$$X_{n_k} < \varepsilon.$$

Since the original sequence is decreasing, for every $n \geq n_{k_0}$,

$$0 \leq X_n \leq X_{n_{k_0}} < \varepsilon.$$

Therefore,

$$X_n \to 0.$$

Hence

$$X_n \xrightarrow{\text{a.s.}} 0.$$

□

9.5 Jensen's Inequality

Theorem 9.5.1 (Jensen's Inequality)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be convex and suppose the relevant expectations exist. Then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

For example, taking $f(x) = |x|$,

$$|\mathbb{E}[X]| \leq \mathbb{E}[|X|].$$

Proof. Let

$$\mu = \mathbb{E}[X].$$

For a differentiable convex function, the tangent line at μ lies below the graph:

$$f(x) \geq f(\mu) + f'(\mu)(x - \mu).$$

Taking expectations,

$$\begin{aligned}\mathbb{E}[f(X)] &\geq \mathbb{E}[f(\mu) + f'(\mu)(X - \mu)] \\ &= f(\mu) + f'(\mu)\mathbb{E}[X - \mu].\end{aligned}$$

Since

$$\mathbb{E}[X - \mu] = 0,$$

we obtain

$$\mathbb{E}[f(X)] \geq f(\mu) = f(\mathbb{E}[X]).$$

□

9.6 An Inequality Between Moments

For $n \geq 1$,

$$\Big(\mathbb{E}[|X|^n]\Big)^{1/n} \leq \Big(\mathbb{E}[|X|^{n+1}]\Big)^{1/(n+1)}.$$

To see this, let

$$Y = |X|^n$$

and define

$$f(y) = y^{1+1/n}, \qquad y \geq 0.$$

Since

$$f''(y) = \left(1 + \frac{1}{n}\right) \frac{1}{n} y^{1/n-1} > 0,$$

the function f is convex. By Jensen's inequality,

$$f(\mathbb{E}[Y]) \leq \mathbb{E}[f(Y)].$$

Therefore,

$$\Big(\mathbb{E}[|X|^n]\Big)^{(n+1)/n} \leq \mathbb{E}[|X|^{n+1}],$$

and hence

$$\Big(\mathbb{E}[|X|^n]\Big)^{1/n} \leq \Big(\mathbb{E}[|X|^{n+1}]\Big)^{1/(n+1)}.$$

In particular, if

$$\mathbb{E}[|X|^{n+1}] < \infty,$$

then

$$\mathbb{E}[|X|^n] < \infty.$$

9.7 Strong Law of Large Numbers

Let $X_1, X_2, \ldots$ be i.i.d. random variables with

$$\mathbb{E}[X_i] = \mu.$$

The strong law of large numbers states that, under suitable integrability assumptions,

$$\frac{1}{n} \sum_{k=1}^{n} X_k \xrightarrow{\text{a.s.}} \mu.$$

In the proof below, we assume the stronger condition

$$\mathbb{E}[|X_1|^4] < \infty.$$

Subtracting the mean, it is enough to prove the result when $\mu = 0$. Define

$$S_n = \sum_{k=1}^{n} X_k.$$

We will estimate $\mathbb{E}[S_n^4]$.

9.8 Fourth-Moment Estimate

Since

$$S_n^2 = \sum_{k=1}^{n} X_k^2 + 2\sum_{i<j} X_i X_j,$$

we have

$$S_n^4 = \left(\sum_{k=1}^{n} X_k^2\right)^2 + 4\left(\sum_{k=1}^{n} X_k^2\right)\left(\sum_{i<j} X_i X_j\right) + 4\left(\sum_{i<j} X_i X_j\right)^2.$$

Since the X_i's are independent and centered, many mixed terms have expectation zero. First,

$$\mathbb{E}[X_i^4] = \mathbb{E}[X_1^4] =: C_0.$$

For $i \neq j$,

$$\begin{aligned}\mathbb{E}[X_i^2 X_j^2] &= \mathbb{E}[X_i^2]\mathbb{E}[X_j^2] \\ &= \left(\mathbb{E}[X_1^2]\right)^2 =: C_1.\end{aligned}$$

Also,

$$\mathbb{E}[X_k^2 X_i X_j] = 0$$

for $i < j$. Indeed, if $k \notin \{i, j\}$, then

$$\mathbb{E}[X_k^2 X_i X_j] = \mathbb{E}[X_k^2]\mathbb{E}[X_i]\mathbb{E}[X_j] = 0.$$

If $k = i$, then

$$\mathbb{E}[X_i^3 X_j] = \mathbb{E}[X_i^3]\mathbb{E}[X_j] = 0,$$

and similarly when $k = j$.
For a product

$$X_i X_j X_{\tilde{i}} X_{\tilde{j}},$$

the expectation is also zero unless the indices occur in pairs. Consequently,

$$\mathbb{E}[S_n^4] \leq nC_0 + 6\binom{n}{2} C_1.$$

Therefore, for some constant $C_2 > 0$,

$$\mathbb{E}[S_n^4] \leq C_2 n^2.$$

9.9 Proof of the Strong Law

Fix $n \geq 1$. By Markov's inequality,

$$\begin{aligned}\mathbb{P}\left(\left|\frac{S_n}{n}\right| > n^{-1/8}\right) &= \mathbb{P}(|S_n| > n^{7/8}) \\ &= \mathbb{P}(|S_n|^4 > n^{7/2}) \\ &\leq \frac{\mathbb{E}[S_n^4]}{n^{7/2}} \\ &\leq C_2 n^{-3/2}.\end{aligned}$$

Hence

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\left|\frac{S_n}{n}\right| > n^{-1/8}\right) \leq C_2 \sum_{n=1}^{\infty} n^{-3/2} < \infty.$$

By the Borel–Cantelli lemma,

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > n^{-1/8} \text{ infinitely often}\right) = 0.$$

Therefore, with probability 1, there exists N such that for every $n \geq N$,

$$\left|\frac{S_n}{n}\right| \leq n^{-1/8}.$$

Since $n^{-1/8} \to 0$,

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} 0.$$

Returning to the case $\mathbb{E}[X_i] = \mu$,

$$\frac{1}{n}\sum_{k=1}^{n} X_k - \mu = \frac{1}{n}\sum_{k=1}^{n}(X_k - \mu) \xrightarrow{\text{a.s.}} 0.$$

Hence

$$\frac{1}{n}\sum_{k=1}^{n} X_k \xrightarrow{\text{a.s.}} \mu.$$

9.10 Dominated Convergence Theorem

The final result in these notes concerns the interchange of limits and expectations.

Theorem 9.10.1 (Dominated Convergence Theorem)**.** Suppose

$$X_n \xrightarrow{\text{a.s.}} X$$

and there exists a random variable Y such that

$$|X_n|^p \leq Y$$

for every $n \geq 1$, where

$$\mathbb{E}[Y] < \infty.$$

Then

$$\mathbb{E}[|X_n - X|^p] \to 0.$$

Acknowledgements

These notes are based primarily on the Probability Theory course taught by Prof. Hongjie Liang. I am grateful to Prof. Liang for his lectures, explanations, and the mathematical perspective developed throughout the course.
I would also like to express my sincere thanks to Yanqi Zeng for sharing her handwritten course notes. Since the original set of notes available to me was incomplete, her notes provided an important source for reconstructing and organizing several parts of the course, especially the material on convergence, the Borel–Cantelli lemma, Jensen's inequality, and the strong law of large numbers.
This document was reorganized and typeset in LaTeX for the purpose of reviewing, preserving, and more conveniently retrieving the material learned in the course. Any errors introduced during the reorganization and typesetting are my own.